Lecture 11:

Data Fitting and Regression

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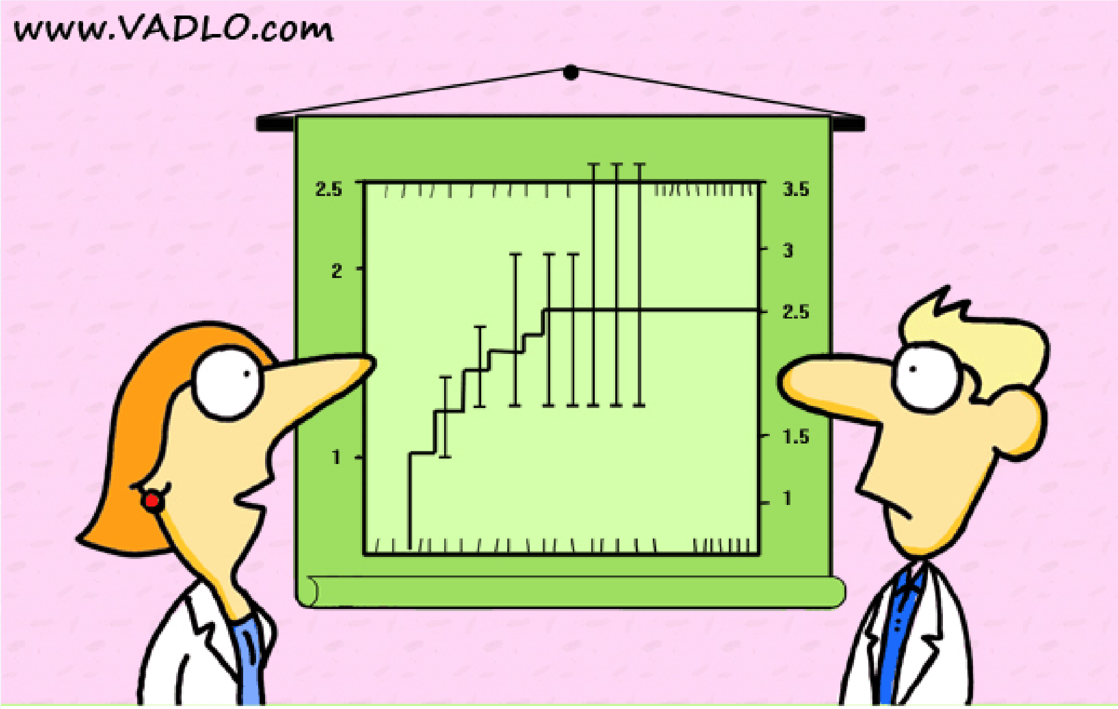
Numerical Methods for Computational Science

Course Map

Week 1:	Introduction to Scientific Computing and Python Libraries
Week 2:	Linear Algebra Fundamentals
Week 3:	Vector Calculus
Week 4:	Numerical Differentiation and Integration
Week 5:	Solving Nonlinear Equations
Week 6:	Probability Theory Basics
Week 7:	Random Variables and Distributions
Week 8:	Statistics for Data Science
Week 9:	Eigenvalues and Eigenvectors
Week 10:	Simulation and Monte Carlo Method
Week 11:	**Data Fitting and Regression**
Week 12:	Optimization Techniques
Week 13:	Machine Learning Fundamentals



"Did you really have to show the error bars?"

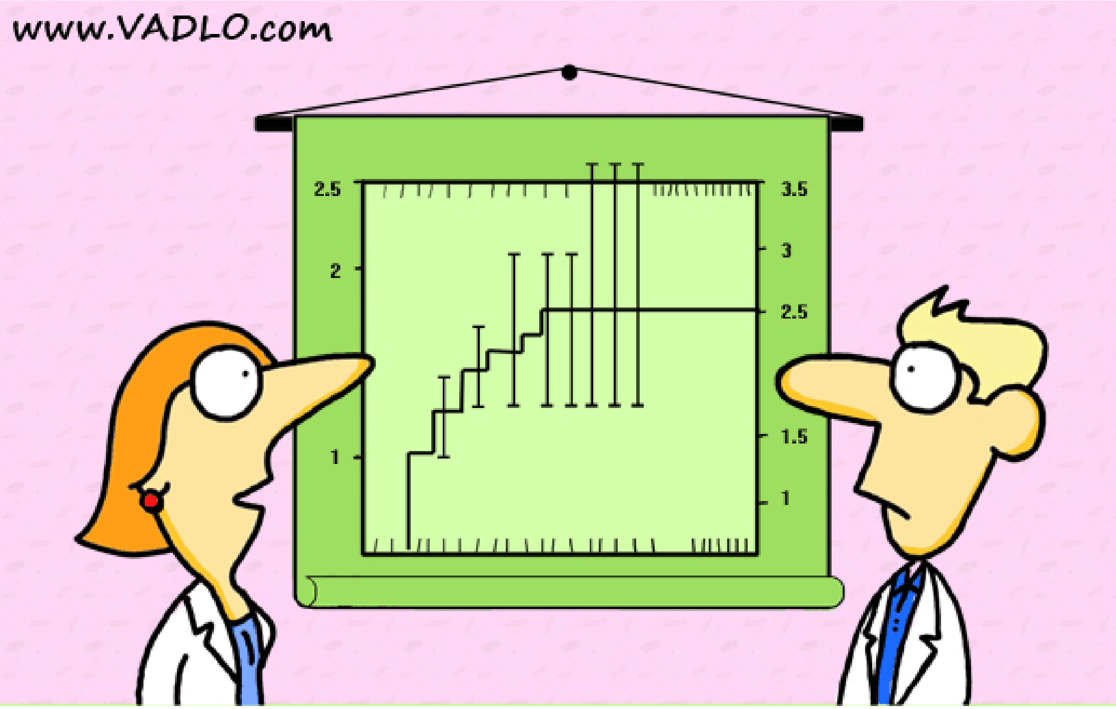
Outline

Linear Regression

- The Math
- What is Linear?
- Stats
- a Python Example

Logistic Regression

Curve Fitting



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Data Fitting and Regression:

The Math
What is Linear?
Stats
a Python Example

problem: a parameter y depends on many **regressors** x_n

goal: finding a model that tells us **how** y depends on each x_n
deriving a model to **predict** y from new data points

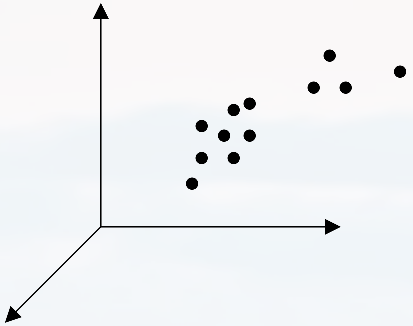
N

k^{th} data point

	x_1	x_2	x_3	x_4	x_5	y_k
Index	molecular_weight	electronegativity	bond_lengths	num_hydrogen_bonds	logP	toxicity_score
0	341.704	2.65585	3.09407	2	9.11147	80.9281
1	335.951	3.22262	2.89039	7	8.92848	83.4911
2	235.203	2.44115	2.48203	1	6.49731	61.8406
3	246.505	2.76656	2.71547	7	7.45089	57.0538
4	437.939	3.4801	3.59569	3	10.9156	131.326

Data Fitting and Regression:

The Math
What is Linear?
Stats
a Python Example



idea: data point y_k in N dimensional space

$$\rightarrow y_k = f(x_1, \dots x_n, \dots x_N) + \epsilon \qquad \text{for each data point } k$$

ansatz:

$$\boxed{y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon}$$

linear combination

y: response
x: regressors **(assumed to be independent)**
β: factors **(how a regressor contributes to the response)**
β_0: intercept
ε: error **(stochasticity of the data, assumed to be normally dist.)**

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k^{th} data point

	x_1	x_2	x_3	x_4	x_5	y_k
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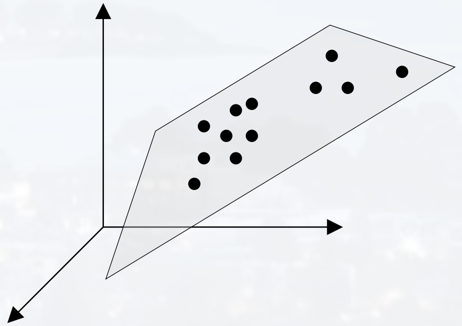


linear ≠ not curved

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n\, x_n^n + \epsilon$$

...is still linear just define: $\bar{x}_n := x_n^n$

$$y_k = \beta_1\, x_n^{\beta_2}$$

...is still linear

just use log: $\bar{y}_k = \log(y_k) = \log(\beta_1) + \beta_2 \log(x_n)$
$= \bar{\beta}_1 + \beta_2 \bar{x}_n$

As long as we can recover the linear structure by any transformation → it is linear

in part. log scaling is quite common examples:

- **log fold change (DESeq/RNASeq)**
- **log odds ratio (comparing models, HMM)**
- **sound → dB is a log unit**
- **log incidence rates (medical studies)**
- **percentiles (medical studies)**
- **.....**

The Math
What is Linear?
Stats
a Python Example

y:	**response**
x:	**regressors**
β:	**factors**
$\boldsymbol{\beta_0}$**:**	**intercept**
ε:	**error**

...what is **not** linear?

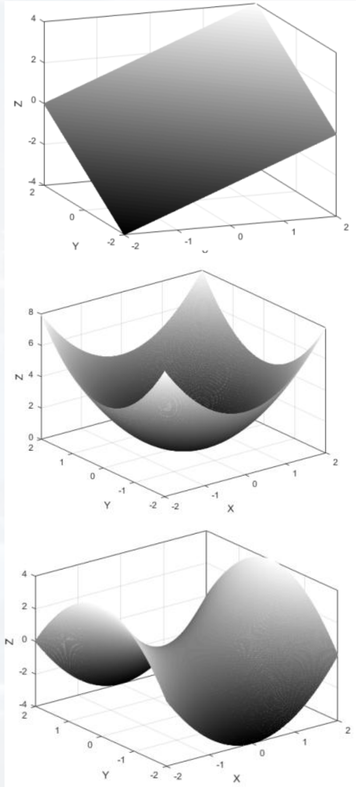
$$y_k = \beta_0 + \beta_1 x_n^{\beta_2}$$ log trick does not work here

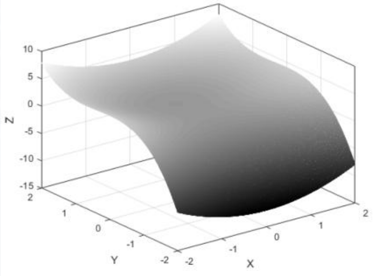
general: linear refers to the **factors**

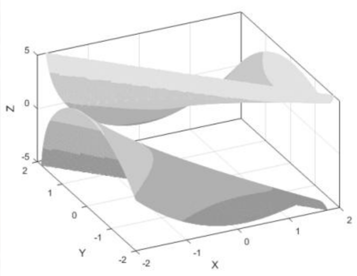
$$y_k = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$ 2*D* plane in 3*D* space

$$y_k = \beta_0 + \beta_1 {x_1}^2 + \beta_2 {x_2}^2$$ 2*D* parabolic

$$y_k = \beta_0 + \beta_1 {x_1}^2 - \beta_2 {x_2}^2$$ 2*D* hyperbolic







...and many more...

all linear

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

y:	**response**
x:	**regressors**
β:	**factors**
$\boldsymbol{\beta_0}$**:**	**intercept**
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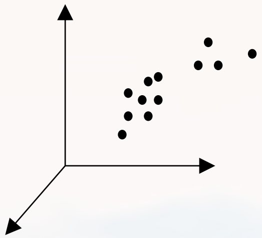
Data Fitting and Regression:



The Math
What is Linear?
Stats
a Python Example

for K data points in N dimensional space

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

$$\underbrace{\begin{pmatrix} y_1 \\ \dots \\ y_k \\ \dots \\ \dots \\ y_K \end{pmatrix}}_{Y} = \underbrace{\begin{pmatrix} 1 & x_{11} & x_{12} \dots x_{1n} \dots x_{1N} \\ \dots & \dots & \dots \\ 1 & x_{k1} & x_{kn} \\ 1 & \dots & \dots \\ 1 & x_{K1} & x_{K2} \dots x_{Kn} \dots x_{KN} \end{pmatrix}}_{X} \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \\ \dots \\ \beta_n \\ \dots \\ \beta_N \end{pmatrix}}_{\beta} + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \dots \\ \varepsilon_k \\ \dots \\ \varepsilon_K \end{pmatrix}}_{\varepsilon}$$

y:	**response**
x:	**regressors**
β:	**factors**
β_0:	**intercept**
ε:	**error**

$$\boxed{Y = X\beta + \varepsilon}$$

N

k^{th} data point

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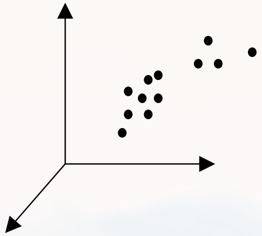
Data Fitting and Regression:



The Math
What is Linear?
Stats
a Python Example

for *K* data points in *N* dimensional space

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

$$\underbrace{\begin{pmatrix} y_1 \\ \dots \\ y_k \\ \dots \\ \dots \\ y_K \end{pmatrix}}_{Y} = \underbrace{\begin{pmatrix} 1 & x_{11} & x_{12} \dots x_{1n} \dots x_{1N} \\ \dots & \dots & \dots \\ 1 & x_{k1} & x_{kn} \\ 1 & \dots & \dots \\ 1 & x_{K1} & x_{K2} \dots x_{Kn} \dots x_{KN} \end{pmatrix}}_{X} \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \\ \dots \\ \beta_n \\ \dots \\ \beta_N \end{pmatrix}}_{\beta} + \underbrace{\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \dots \\ \varepsilon_k \\ \dots \\ \varepsilon_K \end{pmatrix}}_{\varepsilon}$$

$$\boxed{\boldsymbol{Y = X\beta + \varepsilon}}$$

y:	**response**
x:	**regressors**
β:	**factors**
$\boldsymbol{\beta_0}$**:**	**intercept**
ε:	**error**

<u>fitting:</u> finding the best β in terms of minimizing the errors

$$(Y - X\beta)^T (Y - X\beta) = \sum_k \varepsilon_k{}^2$$

the model

$$\frac{\partial}{\partial \beta} \sum_k \varepsilon_k{}^2 = 0 \longrightarrow \beta_{best} = \hat{\beta} = (X^T X)^{-1} X^T Y \longrightarrow \boldsymbol{\hat{Y} = X\hat{\beta}} = X(X^T X)^{-1} X^T Y$$

X and Y are all observables

$\boxed{\boldsymbol{Y = X\beta + \varepsilon}}$ finding the best β in terms of minimizing the errors

$$(Y - X\beta)^T(Y - X\beta) = \sum_k {\varepsilon_k}^2$$

the model $\hat{Y} = X\hat{\beta} = \underbrace{X\left(X^TX\right)^{-1}X^T}_{\text{hat matrix } H}Y$

y:	**response**
x:	**regressors**
β:	**factors**
$\boldsymbol{\beta_0}$**:**	**intercept**
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some properties of the hat matrix:

- $H = H^T$ (symmetry)
- $HH = H \rightarrow H^n = H$ (idempotency)

evaluating the fit:

$$\hat{\varepsilon} = Y - X\hat{\beta} = Y - \hat{Y} = (I - H)Y$$

$$\hat{\varepsilon}^T\hat{\varepsilon} = [(I-H)Y]^T(I-H)Y = Y^T(I-H)^T(I-H)Y = Y^T(I-H)Y$$ sum of squared errors (SSE)

$$\hat{\sigma}^2 = \frac{\hat{\varepsilon}^T\hat{\varepsilon}}{K - N}$$ ← **degrees of freedom**

variance or mean of squared errors (MSE)

Data Fitting and Regression:

y:	**response**
x:	**regressors**
β:	**factors**
$\boldsymbol{\beta_0}$**:**	**intercept**
ε:	**error**

summary:

the model: $Y = X\beta + \varepsilon$

the fit: $\hat{Y} = X\hat{\beta} = X(X^TX)^{-1}X^TY$

SSE: $\hat{\varepsilon}^T\hat{\varepsilon} = Y^T(I - H)Y$ (after the fit)

variance or MSE: $\hat{\sigma}^2 = \frac{\hat{\varepsilon}^T\hat{\varepsilon}}{K-N}$ (after the fit)

often fit quality is judged by $R^2 := 1 - \frac{\sum_k(\hat{y}_k - y_k)^2}{\sum_k(y_k - \langle y\rangle)^2}$

or adjusted R^2 $\bar{R}^2 := R^2 - (1 - R^2)\frac{N}{K-N-1}$

and it is said that the fit is good if R^2 is close to one....

...but that is not true...

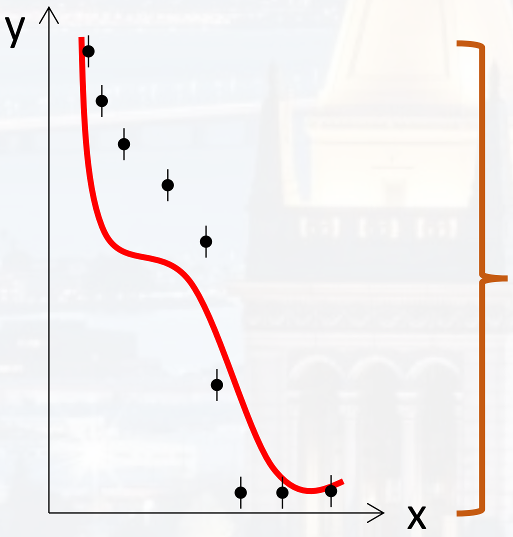
Data Fitting and Regression:

$$R^2 = 1 - \frac{\boxed{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}}{\boxed{\sum_{i=1}^{N}(y_i - \bar{y})^2}}$$

variance data vs model
(aka residual sum of squares)

variance of the data
(aka total sum of squares)

Note: do not confuse R^2 with Pearsons coefficient: $\rho = \frac{cov(x,y)}{\sqrt{var(x)var(y)}}$



data variance can be huge
(i. e. exponential functions)
→ R^2 could be around 1.0
even if fit is completely off!

$\bar{y}$: mean of the data point values

$\bar{y}$:	**mean of the data point values**
y_i:	**measured value of data point**
σ_i:	**statistical error of y_i (often aka ey_i)**
$\hat{y}_i$:	**prediction by the model *after the fit***
N :	**number of data points**
p:	**number of fit parameter**

Data Fitting and Regression:

$$R^2 = 1 - \frac{\boxed{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}}{\boxed{\sum_{i=1}^{N}(y_i - \bar{y})^2}}$$

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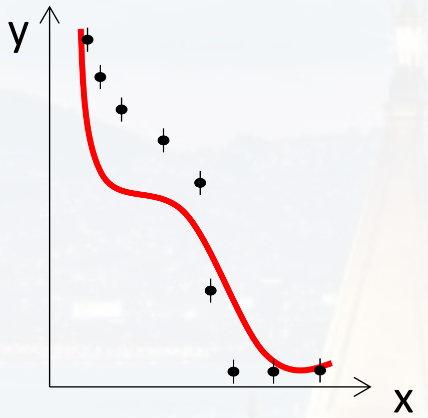

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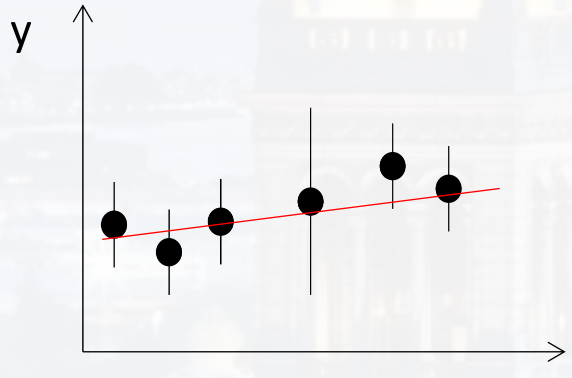

$$\frac{\text{variance data vs model (aka residual sum of squares)}}{\text{variance of the data (aka total sum of squares)}} \approx 1 \rightarrow R^2 = 0$$

although the fit is good!

$\bar{y}$:	**mean of the data point values**
y_i:	**measured value of data point**
σ_i:	**statistical error of y_i (often aka ey_i)**
$\hat{y}_i$:	**prediction by the model *after the fit***
N :	**number of data points**
p:	**number of fit parameter**

Data Fitting and Regression:

$$R^2 = 1 - \frac{\boxed{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}}{\boxed{\sum_{i=1}^{N}(y_i - \bar{y})^2}}$$

variance data vs model
(aka residual sum of squares)

variance of the data
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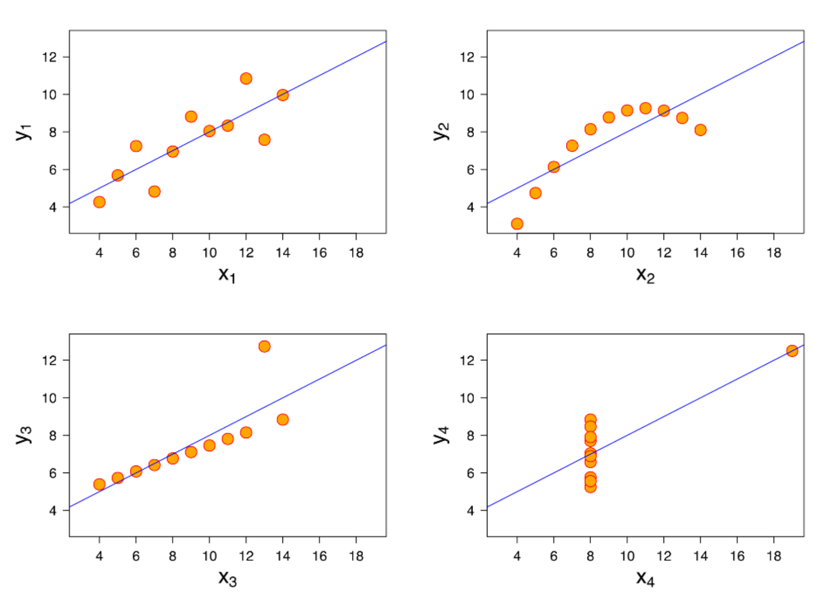

all plots: same R^2

conclusion:

R^2 is **not** a measure of the fit quality (but χ^2 is)

Given a good fit, R^2 tells how strong the dependent variable responds to the independent variable

Also, Wiki is full of examples…
…and warnings (see “caveats” therein)

- The Math
- What is Linear?
- Stats
- **a Python Example**

see `Walk_Through_LinRegression.ipynb`

```python
import numpy as np

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import pylab
import scipy.stats as stats
import statsmodels.api as sm

from statsmodels.formula.api import ols
from sklearn.preprocessing import MinMaxScaler
```

- pandas: reading .xlsx, .csv, .txt …
- matplotlib.pyplot: standard plots
- seaborn: fancy plots: here a pair-plot
- pylab, scipy.stats: Q-Q plot
- statsmodels.api, statsmodels.formula.api: the actual super tool for superb data analysis
- sklearn.preprocessing: scaling and normalizing

1) loading data
2) plotting data
3) scaling data
4) fitting model
5) evaluating model

```
Train  = pd.read_csv("molecular_train_gbc.csv")
Test   = pd.read_csv("molecular_test_gbc.csv")
```

	x_1	x_2	x_3	x_4	x_5	y_k
Index	molecular_weight	electronegativity	bond_lengths	num_hydrogen_bonds	logP	toxicity_score
0	341.704	2.65585	3.09407	2	9.11147	80.9281
1	335.951	3.22262	2.89039	7	8.92848	83.4911
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$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

y: toxicity_score

x_n : molecular_weight, electronegativity, bond_lengths, num_hydrogen_bonds, logP

1) loading data
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```
Train  = pd.read_csv("molecular_train_gbc.csv")
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out = sns.pairplot(Train, kind = "kde", \
                   plot_kws = {'color':[176/255, 224/255, 230/255]},\
                   diag_kws = {'color': 'black'})
out.map_offdiag(plt.scatter, color = 'black')
```

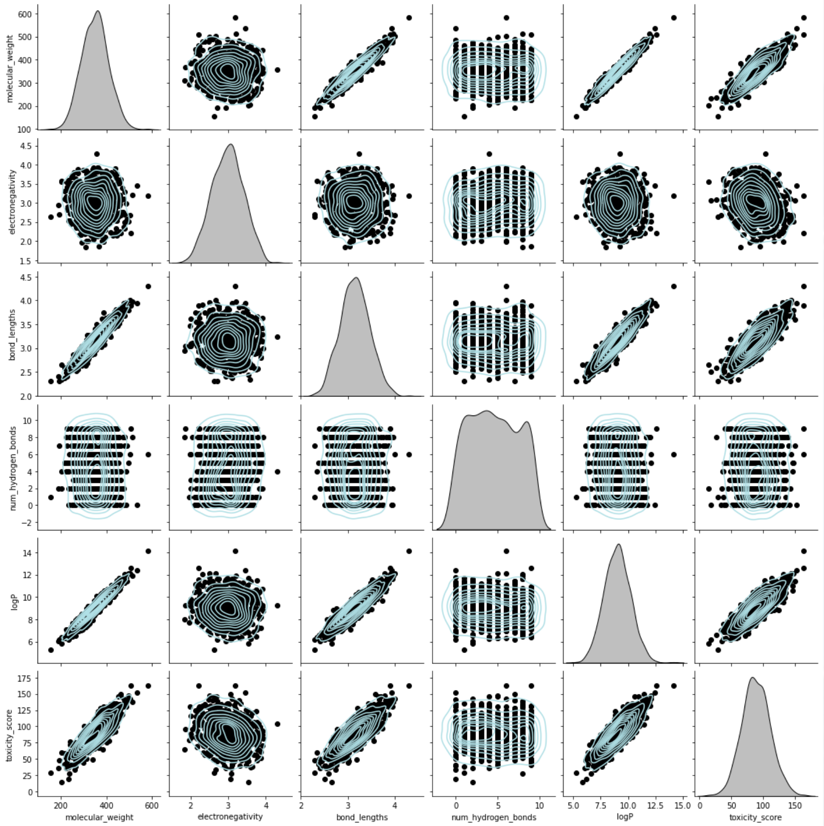

Data Fitting and Regression:


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out.map_offdiag(plt.scatter, color = 'black')
```

```
scaler  = MinMaxScaler(feature_range = (0, 1))
TrainS  = scaler.fit_transform(Train)
TestS   = scaler.transform(Test)
```

the scaler returns an `np.array`
→ convert back to `data frame`

```
TrainS = pd.DataFrame(TrainS, columns = Train.columns)
TestS  = pd.DataFrame(TestS, columns = Train.columns)
```

1) loading data
2) plotting data
3) scaling data
4) fitting model
5) evaluating model

```
TrainS = pd.DataFrame(TrainS, columns = Train.columns)
TestS  = pd.DataFrame(TestS,  columns = Train.columns)
```

```
equation = 'toxicity_score ~ ' + '+'.join(Train.columns[:-1])
print(equation)
```

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

```
toxicity_score ~      molecular_weight + electronegativity +
                      bond_lengths + num_hydrogen_bonds + logP
```

```
my_model = ols(equation, data = TrainS).fit()
my_model.summary()
```

OLS (ordinary least squares)

Data Fitting and Regression:

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

```
my_model.summary()
```

not the fit quality!

```
                            OLS Regression Results
==============================================================================
Dep. Variable:         toxicity_score   R-squared:                       0.790
Model:                            OLS   Adj. R-squared:                  0.789
Method:                 Least Squares   F-statistic:                     597.5
Date:                Fri, 13 Sep 2024   Prob (F-statistic):          3.34e-266
Time:                        20:57:10   Log-Likelihood:                 1013.0
No. Observations:                 800   AIC:                            -2014.
Df Residuals:                     794   BIC:                            -1986.
Df Model:                           5
Covariance Type:            nonrobust
=====================================================================================
                        coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------
Intercept             0.1494      0.012     12.533      0.000       0.126       0.173
molecular_weight      0.7961      0.089      8.982      0.000       0.622       0.970
electronegativity    -0.1682      0.015    -11.591      0.000      -0.197      -0.140
bond_lengths          0.0204      0.049      0.417      0.677      -0.076       0.116
num_hydrogen_bonds    0.0035      0.008      0.458      0.647      -0.011       0.018
logP                  0.1246      0.072      1.723      0.085      -0.017       0.267
==============================================================================
Omnibus:                        2.249   Durbin-Watson:                   1.984
Prob(Omnibus):                  0.325   Jarque-Bera (JB):                2.240
Skew:                          -0.129   Prob(JB):                        0.326
Kurtosis:                       2.980   Cond. No.                         65.6
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

number of data points is much larger than the number of regressors
→ degree of freedom approx. no of obs

p-value for constant model

p-values for factors

2σ conf range of factors

more accurate: determining **the p-values for the factors using ANOVA** for the corresponding residuals

```
table    = sm.stats.anova_lm(my_model, typ = 1)
print(table)
```

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

	df	sum_sq	mean_sq	F	PR(>F)
molecular_weight	1.0	13.346285	13.346285	2847.525516	8.024085e-265
electronegativity	1.0	0.640388	0.640388	136.631363	3.085962e-29
bond_lengths	1.0	0.000684	0.000684	0.145954	7.025342e-01
num_hydrogen_bonds	1.0	0.000703	0.000703	0.150055	6.985866e-01
logP	1.0	0.013917	0.013917	2.969353	8.524510e-02
Residual	794.0	3.721459	0.004687	NaN	NaN

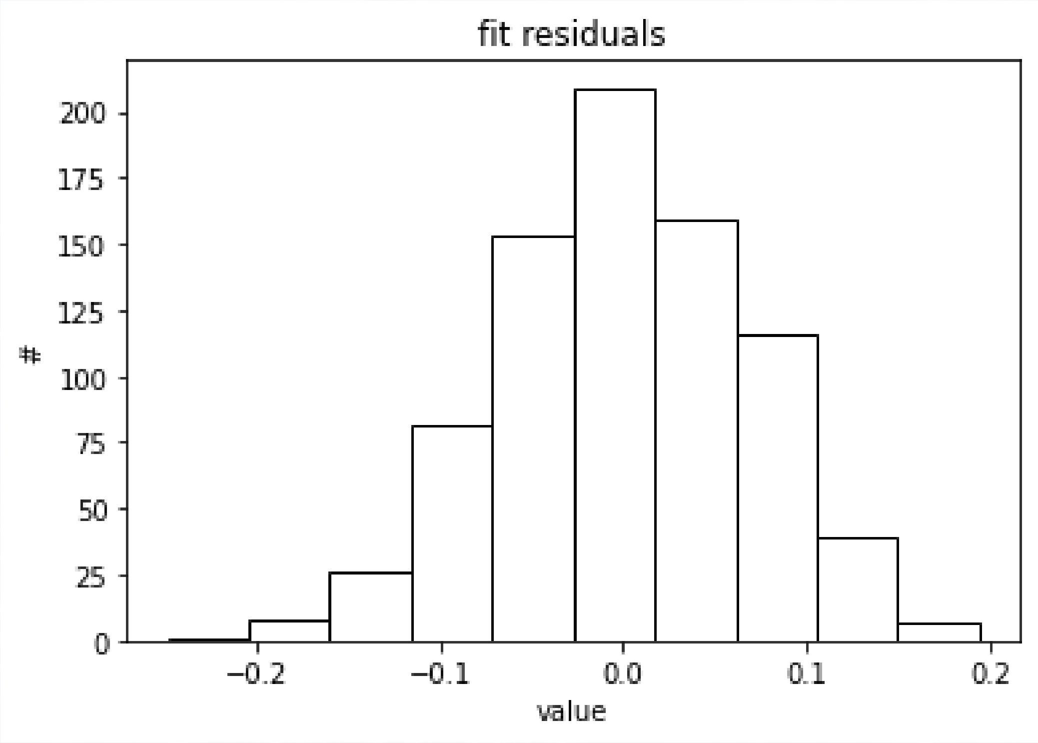
vs from t-test

0.0000
0.0000
0.6766
0.6473
0.0852

```
residuals = my_model.resid

plt.hist(residuals, color = 'w', edgecolor = 'black')
plt.title('fit residuals')
plt.ylabel('#')
plt.xlabel('value')
plt.show()
```

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$



residuals approx. normally distributed around μ = 0

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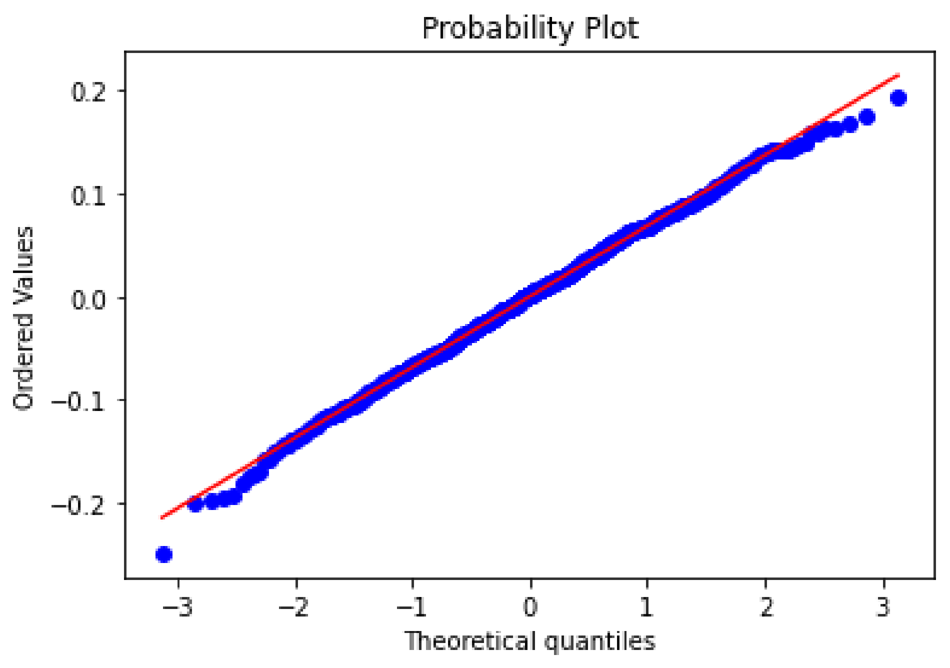
residuals approx. normally distributed around μ = 0

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

```
stats.probplot(residuals, dist = "norm", plot = pylab)
pylab.show()
```



Data Fitting and Regression:

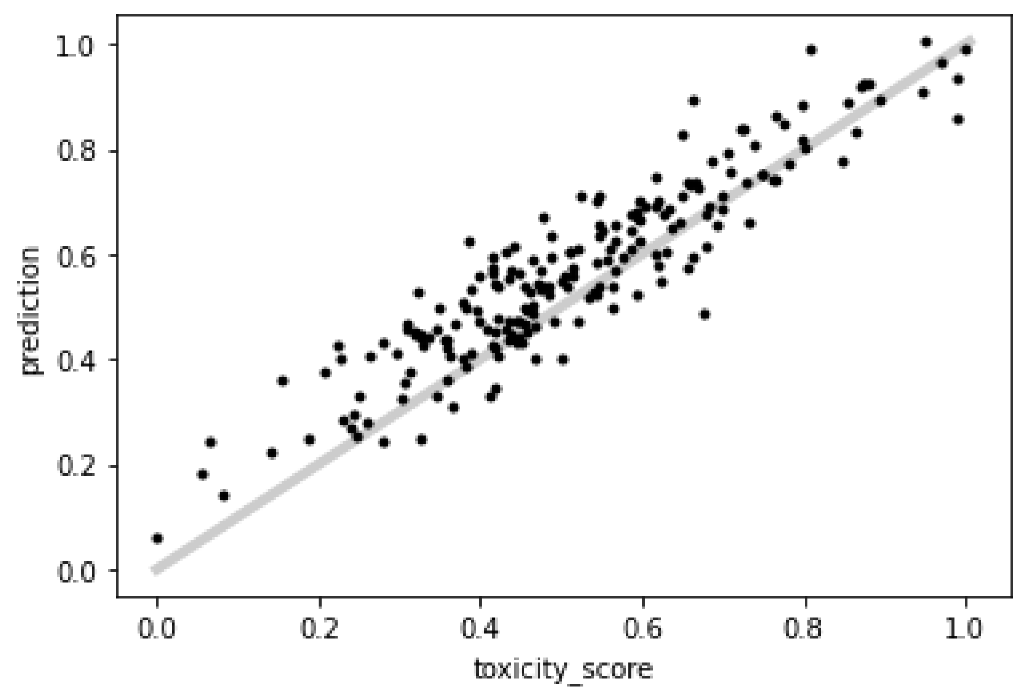

```
Ypred  = my_model.predict(TestS)

higher = np.max([Ypred, TestS.toxicity_score])
lower  = np.min([Ypred, TestS.toxicity_score])

plt.plot([lower, higher], [lower, higher], c = [0, 0, 0, 0.2],\
         linewidth = 4)
plt.scatter(TestS.toxicity_score, Ypred, marker = '.', c = 'k')
plt.ylabel('prediction')
plt.xlabel('toxicity score')
plt.show()
```

1) loading data
2) plotting data
3) scaling data
4) fitting model
5) evaluating model

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$



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```
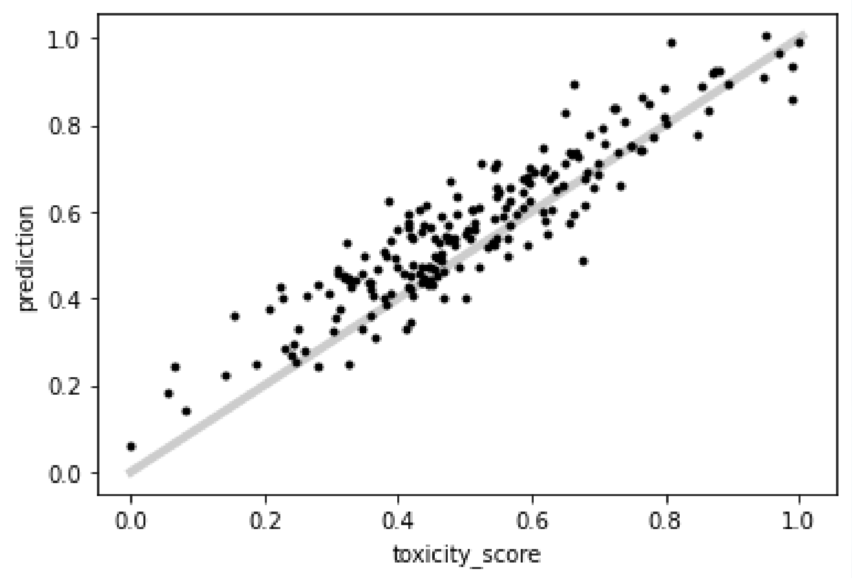
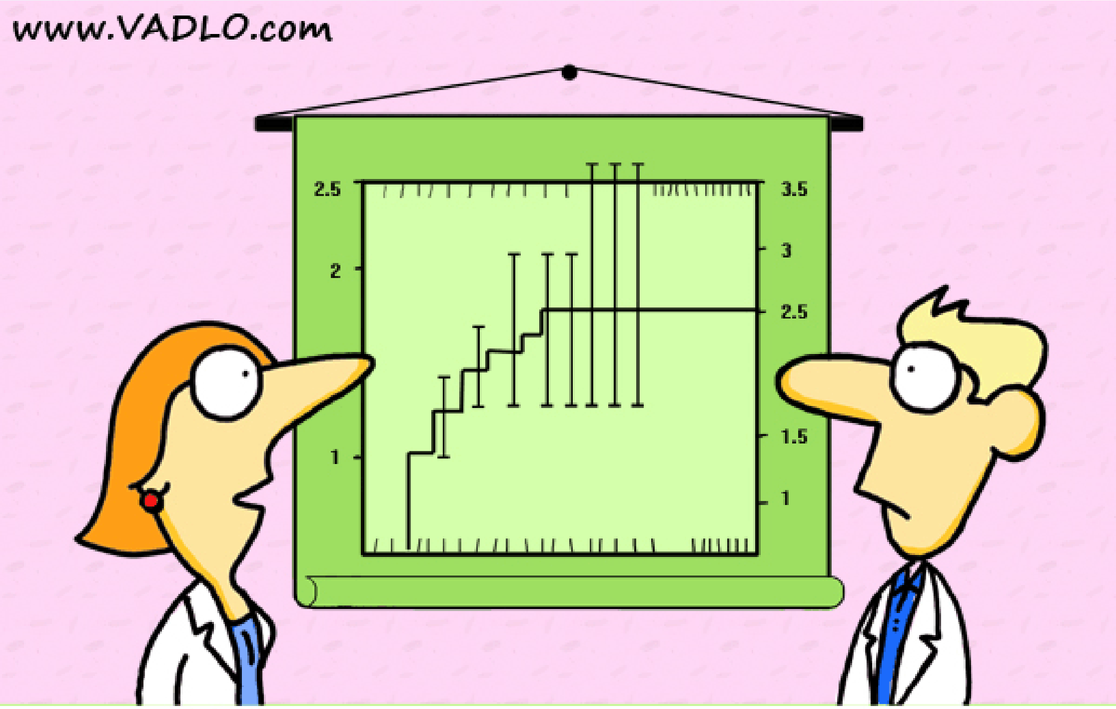




```
mean_dev = np.sum( abs(TestS.toxicity_score - Ypred) )/len(Ypred)
print(mean_dev)
```

5%

1) loading data
2) plotting data
3) scaling data
4) fitting model
5) evaluating model

$$y_k = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$



"Did you really have to show the error bars?"

Outline

Linear Regression

- The Math
- What is Linear?
- Stats
- a Python Example

Logistic Regression

Curve Fitting

Data Fitting and Regression:

<u>linear model:</u> regressors are continuous or categorical, response is continuous

<u>logistic model:</u> response is **categorical**

y:	response
x:	regressors **(assumed to be independent)**
β:	factors
β_0:	intercept
ε:	error **(stochasticity of the data, assumed to be normally dist.)**

Index	molecular_weight	electronegativity	bond_lengths	num_hydrogen_bonds	logP	label
0	341.704	2.65585	3.09407	2	9.11147	Toxic
1	335.951	3.22262	2.89039	7	8.92848	Toxic
2	235.203	2.44115	2.48203	1	6.49731	Non-Toxic
3	246.505	2.76656	2.71547	7	7.45089	Non-Toxic
4	437.939	3.4801	3.59569	3	10.9156	Non-Toxic

Data Fitting and Regression:

y:	response
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ε:	error **(stochasticity of the data, assumed to be normally dist.)**

linear model: regressors are continuous or categorical, response is continuous

logistic model: response is **categorical**

dichotomic model:

probability to be in state A) → ***p***

probability to be in state B) → ***1 - p***

label
Toxic
Toxic
Non-Toxic
Non-Toxic
Non-Toxic

ansatz:

$$\boxed{log\left(\frac{p}{1-p}\right) = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon}$$

log odds ratio: linear model

Data Fitting and Regression: Logistic Regression

dichotomic model:

label
Toxic
Toxic
Non-Toxic
Non-Toxic
Non-Toxic

probability to be in state A) → ***p***

probability to be in state B) → ***1 - p***

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<u>ansatz:</u>

$$\boxed{log\left(\frac{p}{1-p}\right) = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon}$$

log odds ratio: linear model

→ probability for being in a certain state

$$\boxed{p = \frac{e^{\beta_0+\beta_1 x_1+\cdots}}{1+e^{\beta_0+\beta_1 x_1+\cdots}}}$$

often:

$$\boxed{logit(p) = log\left(\frac{p}{1-p}\right)}$$

<u>examples:</u>

- probability that a gene has been mutated
- probability of being diseased (cancer, alzheimer etc) as function of age, environmental influence etc ...
- Verhulst equation: $N(t) = N_0 \frac{e^{rt}}{C+e^{rt}}$
- activation functions in ANNs

Data Fitting and Regression:

$$log\left(\frac{p}{1-p}\right) = \beta_0 + \sum_{n=1}^{N} \beta_n x_n + \epsilon$$

y:	response
x:	regressors **(assumed to be independent)**
β:	factors
β_0:	intercept
ε:	error **(stochasticity of the data, assumed to be normally dist.)**

Note: one can derive the logit function from max. entropy too!

$$\boxed{p = \frac{e^{\beta_0+\beta_1 x_1+\cdots}}{1+e^{\beta_0+\beta_1 x_1+\cdots}} = \frac{1}{1+e^{-\beta_0-\beta_1 x_1-\cdots}}}$$

onset of Alzheimer's disease (AD) is a function of ***age*** and years spent in ***education***
(and other risk factors we ignore here for the sake of simplicity)

education: $\mathrm{d} = x_1$ [yrs]
age: $a = x_2$ [yrs]

model: $p_{AD} = \frac{1}{1+e^{-\beta_0-\beta_1 d-\beta_2 a}}$

+ data set + fit → $\beta_0 = +0.1$
$\beta_1 = -1.5$
$\beta_2 = +0.12$

- *positive* value → *increasing p*
- *negative* value → *decreasing p*
- intercept: "background" prevalence, not related to environmental/internal conditions

Data Fitting and Regression:

y:	response
x:	regressors **(assumed to be independent)**
β:	factors
β_0:	intercept
ε:	error **(stochasticity of the data, assumed to be normally dist.)**

model: $p_{AD} = \frac{1}{1+e^{-\beta_0-\beta_1 d-\beta_2 a}}$

education: $\mathrm{d} = x_1$ [yrs]
age: $a = x_2$ [yrs]

$\beta_0 = +0.1$
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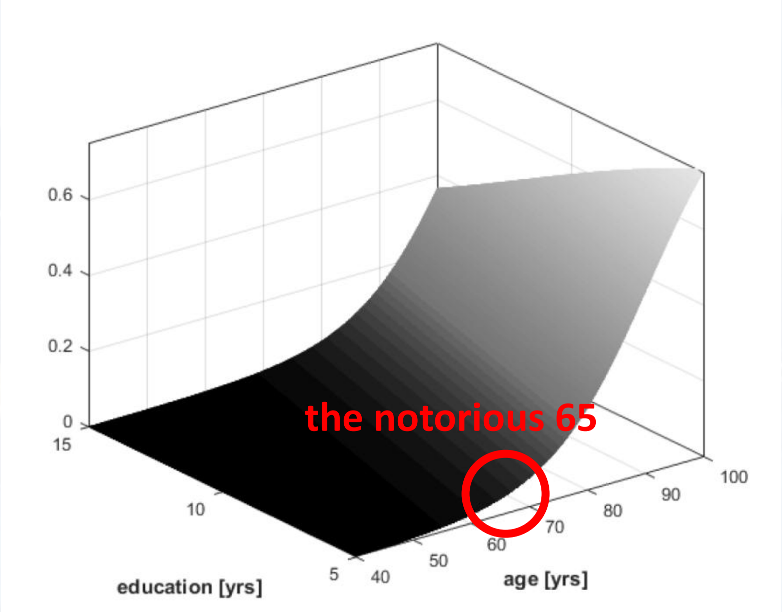

example: ***65yrs*** old person, ***8yrs*** spent in education
→ $p_{AD} = 1.6\%$

65yrs old person, ***13yrs*** spent in education
→ $p_{AD} = 0.001\%$

<u>How does education compensate aging?</u>

$$p_{AD}(d+\bar{d}, a+\bar{a}) = p_{AD}(d, a)$$

→ $\bar{a} = 12.5\,\bar{d}$

hence, one more year prolonged education compensates 12.5 years of aging
(warning: don't confuse correlation with causation here!)

model: $p_{AD} = \frac{1}{1+e^{-\beta_0-\beta_1 d-\beta_2 a}}$

y:	response
x:	regressors **(assumed to be independent)**
β:	factors
β_0:	intercept
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education: $\mathrm{d} = x_1$ [yrs]

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$\beta_0 = +0.1$

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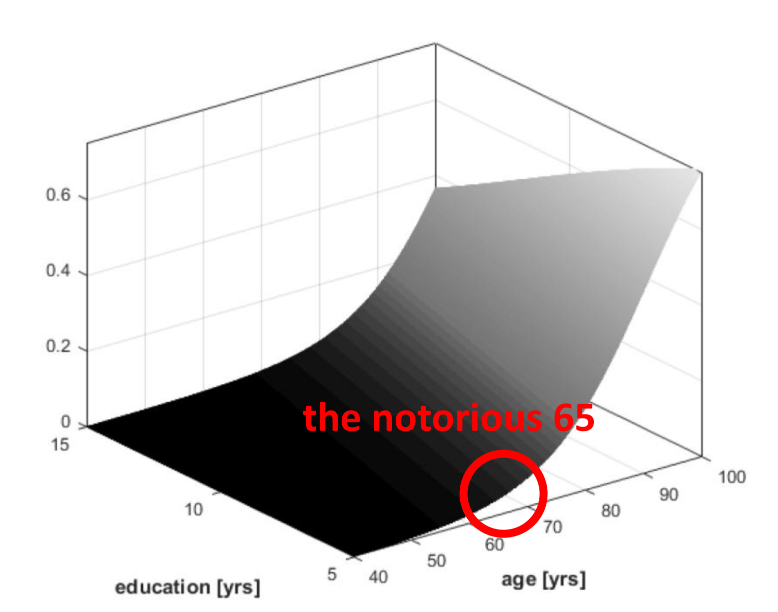

<u>How does the risk of onset changes *per year*?</u>

relative change:

$$\frac{p_{AD}(a+1) - p_{AD}(a)}{p_{AD}(a)} \approx e^{\beta_2} - 1 \approx 12.7\%$$

$p_{AD} \ll 1$ (hence, for small Δa and "young" ages, i. e. below $\approx$ 80yrs)

the risk of getting AD increases by 12.7% every year

(warning: does not mean that it increases by 127% in ten yrs – we made an approximation!)

y:	response
x:	regressors **(assumed to be independent)**
β:	factors
β_0:	intercept
ε:	error **(stochasticity of the data, assumed to be normally dist.)**

model: $$p_{AD} = \frac{1}{1+e^{-\beta_0-\beta_1 d-\beta_2 a}}$$

education: $d = x_1$ [yrs]

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$\beta_0 = +0.1$

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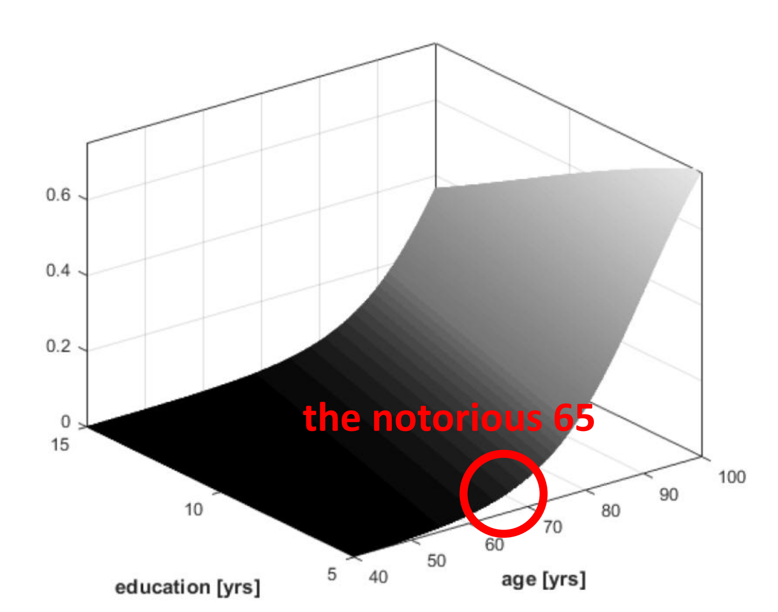
$\beta_2 = +0.12$



<u>How does the risk of onset changes *per year*?</u>

more precise: relative change of the odds ratio

$$\frac{\frac{\partial}{\partial x_i}\left(\frac{p_{AD}}{1-p_{AD}}\right)}{\frac{p_{AD}}{1-p_{AD}}} = \beta_i$$

x_i is the desired regressor, for example, age again (x_2)

the factors β_i indicate how strong (and in which direction) *p* changes wrt a regressor x_i

let us return to the molecule data set:

```
Train   = pd.read_csv("molecular_train_gbc_cat.csv")
Test    = pd.read_csv("molecular_test_gbc_cat.csv")
```

Index	molecular_weight	electronegativity	bond_lengths	num_hydrogen_bonds	logP	label
0	341.704	2.65585	3.09407	2	9.11147	Toxic
1	335.951	3.22262	2.89039	7	8.92848	Toxic
2	235.203	2.44115	2.48203	1	6.49731	Non-Toxic
3	246.505	2.76656	2.71547	7	7.45089	Non-Toxic
4	437.939	3.4801	3.59569	3	10.9156	Non-Toxic

```
import statsmodels.api as sm
```

Data Fitting and Regression:

it is the same data set → plotting and scaling is as before

```
X = sm.add_constant(TrainS)
```

adding the intercept

```
Y = pd.get_dummies(Train['label'])
```

Python needs True/False as categorical

$$p = \frac{e^{\beta_0+\beta_1 x_1+\cdots}}{1+e^{\beta_0+\beta_1 x_1+\cdots}}$$

```
In [48]: print(Y)
   Non-Toxic  Toxic
0      False   True
1      False   True
2       True  False
3       True  False
4       True  False
```

we have two states: toxic / non-toxic

```
my_model = sm.GLM(Y, X, family = sm.families.Binomial()).fit()

my_model.summary()
```

GLM: general linear model

Data Fitting and Regression:

it is the same data set → plotting and scaling is as before

```
X = sm.add_constant(TrainS)
Y = pd.get_dummies(Train['label'])

my_model = sm.GLM(Y, X, family = sm.families.Binomial()).fit()
my_model.summary()
```

```
                 Generalized Linear Model Regression Results
==============================================================================
Dep. Variable:        ['Non-Toxic', 'Toxic']   No. Observations:          800
Model:                                   GLM   Df Residuals:              794
Model Family:                       Binomial   Df Model:                    5
Link Function:                         Logit   Scale:                  1.0000
Method:                                 IRLS   Log-Likelihood:        -332.82
Date:                       Sat, 14 Sep 2024   Deviance:               665.64
Time:                               20:59:18   Pearson chi2:         1.14e+03
No. Iterations:                            6   Pseudo R-squ. (CS):     0.4243
Covariance Type:                   nonrobust
======================================================================================
                         coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------------
const                  6.1641      0.585     10.536      0.000       5.017       7.311
molecular_weight     -10.4920      3.626     -2.893      0.004     -17.599      -3.385
electronegativity      3.2874      0.599      5.492      0.000       2.114       4.461
bond_lengths           0.6736      1.913      0.352      0.725      -3.075       4.422
num_hydrogen_bonds    -0.3082      0.303     -1.018      0.309      -0.902       0.285
logP                  -7.6090      2.978     -2.555      0.011     -13.447      -1.771
======================================================================================
```

$$p = \frac{e^{\beta_0+\beta_1 x_1+\cdots}}{1+e^{\beta_0+\beta_1 x_1+\cdots}}$$

p-value for constant model

p-values for factors

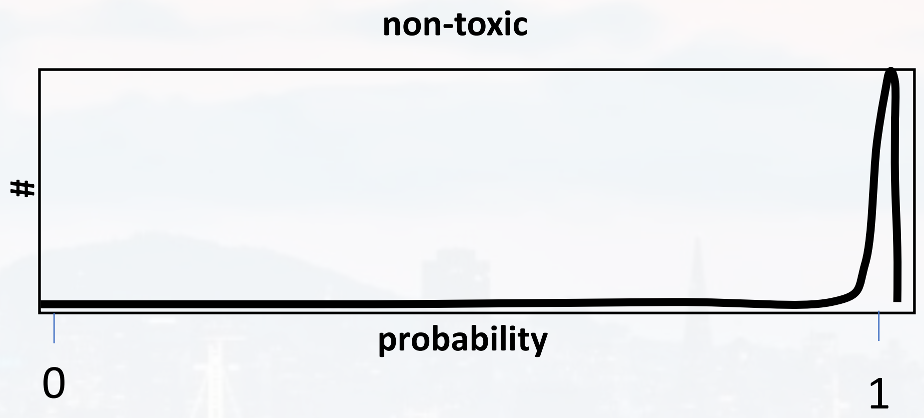
2σ conf range of factors

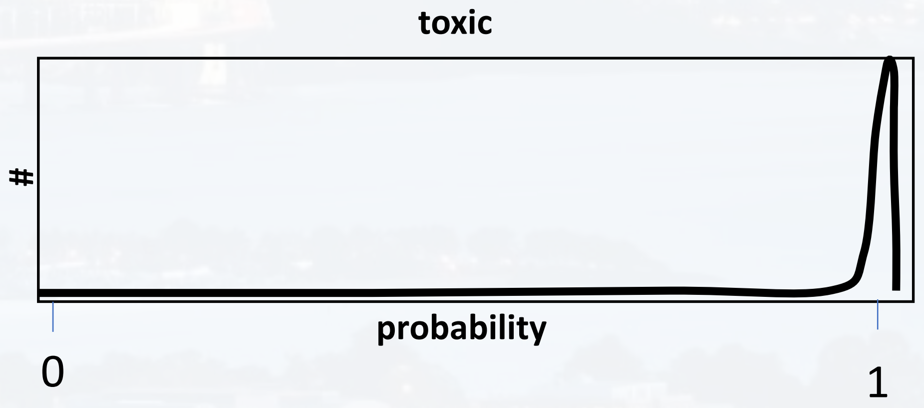
Data Fitting and Regression: Logistic Regression

accuracy: How ***often*** did the model make the correct prediction.
cross-entropy: How ***certain*** was the model when making the prediction.

<u>ideal world:</u>

true label \ predicted label	non-toxic	toxic
non-toxic	**100%**	**0%**
toxic	**0%**	**100%**

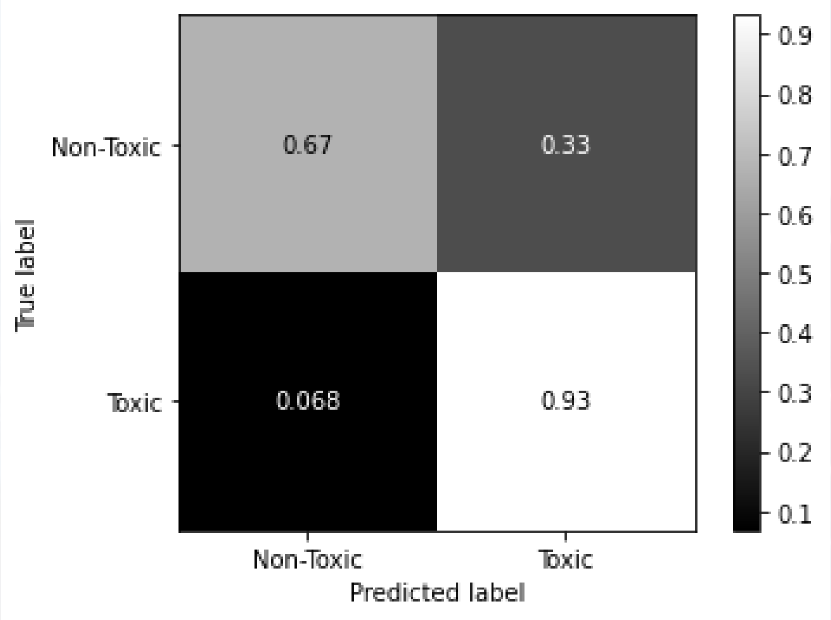


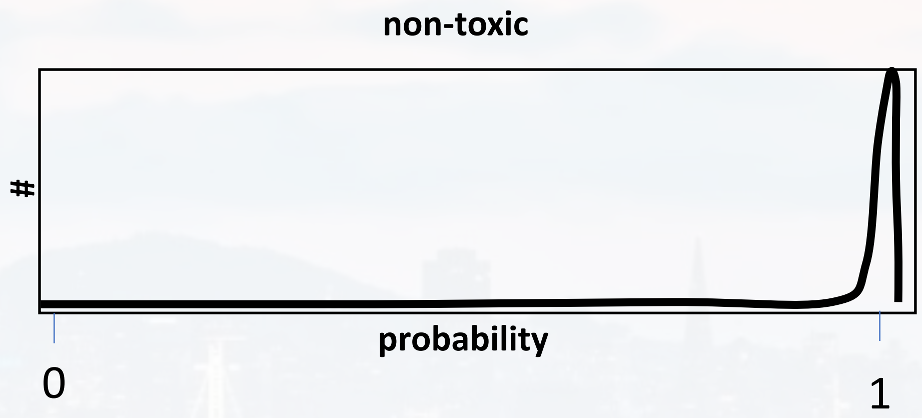


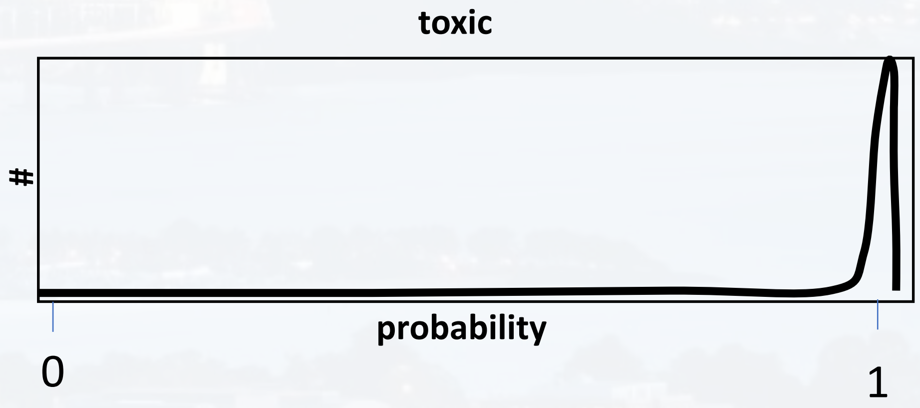
more details see **Chem 277**

accuracy: How ***often*** did the model make the correct prediction.
cross-entropy: How ***certain*** was the model when making the prediction.

<u>real world:</u>







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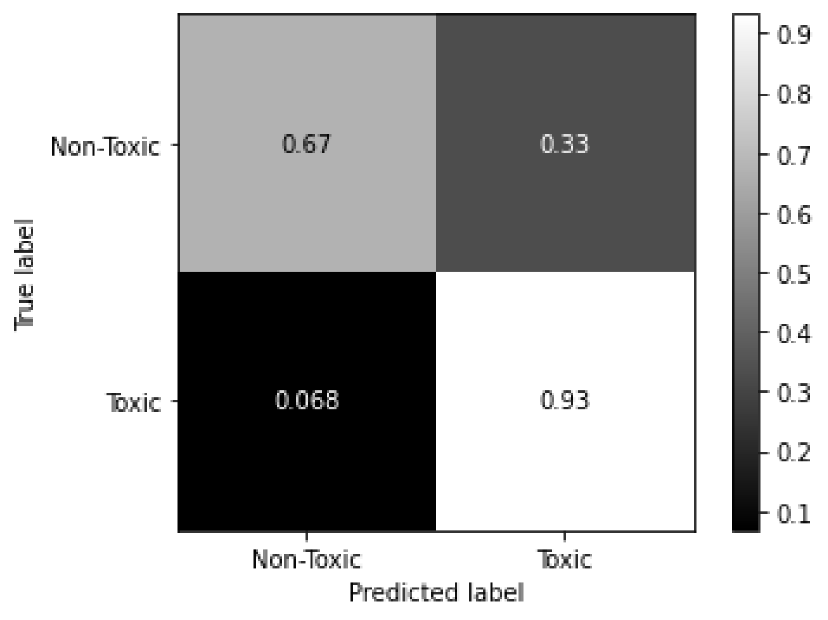
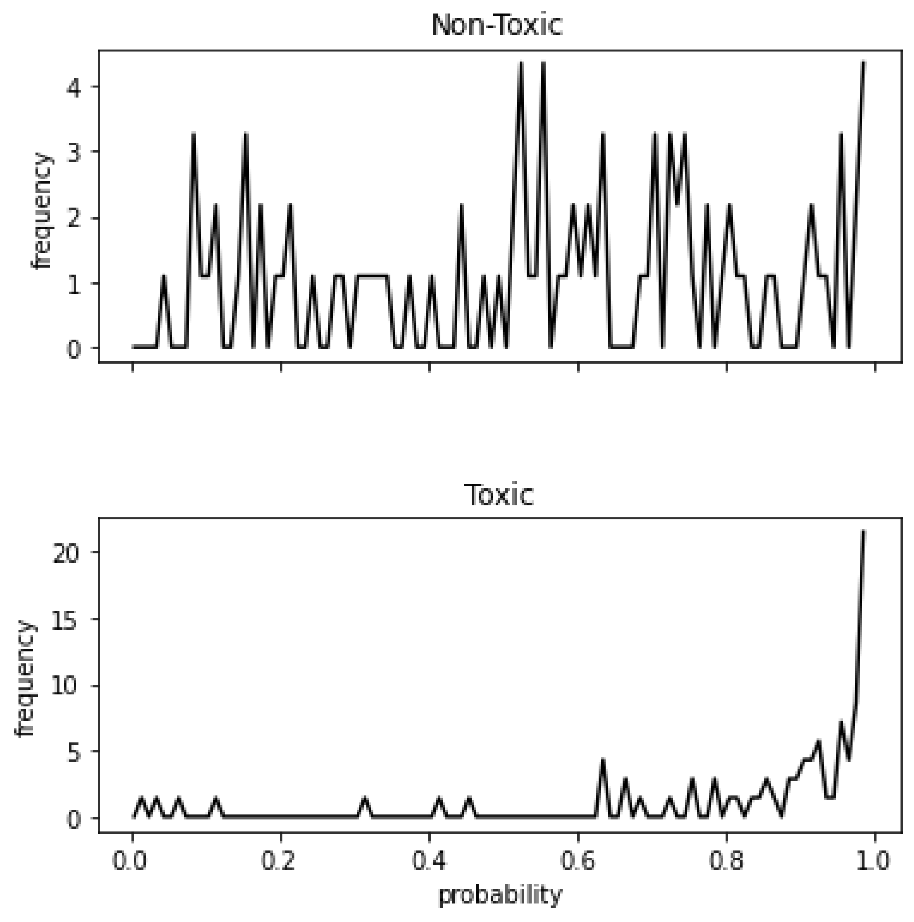
real world:





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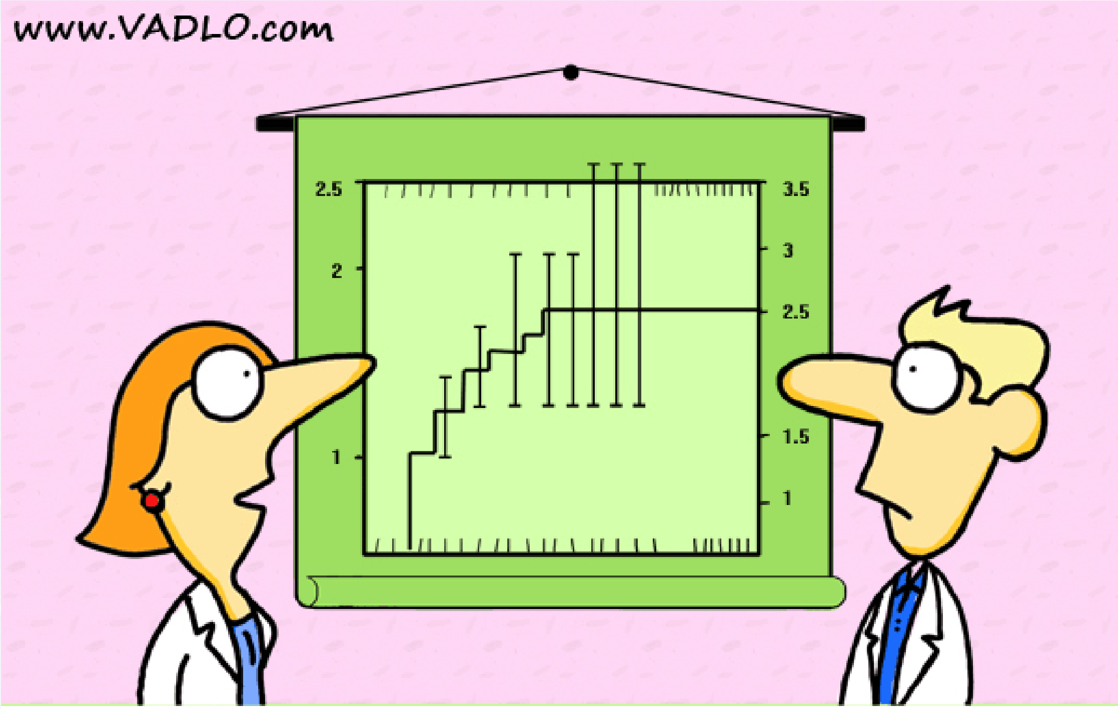

"Did you really have to show the error bars?"

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Linear Regression

- The Math

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- Stats

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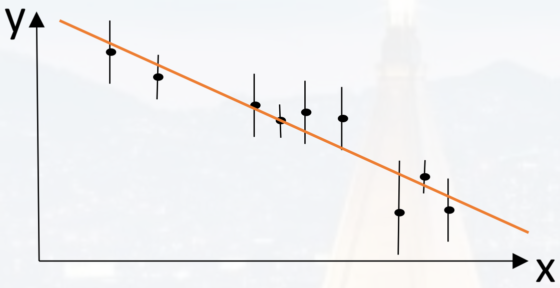
Curve Fitting

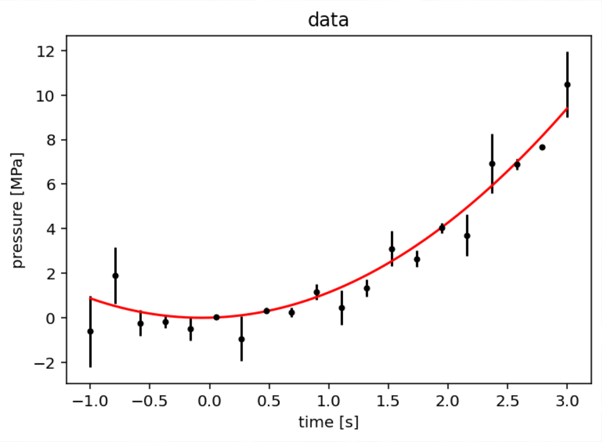
Data Fitting and Regression:

problem: set of data points $y_i = f(x_i)$, ideally each data point y_i has an error σ_i
we know $f(x_i)$

goal: find parameters of $f(x_i)$
once we know the parameters → prediction



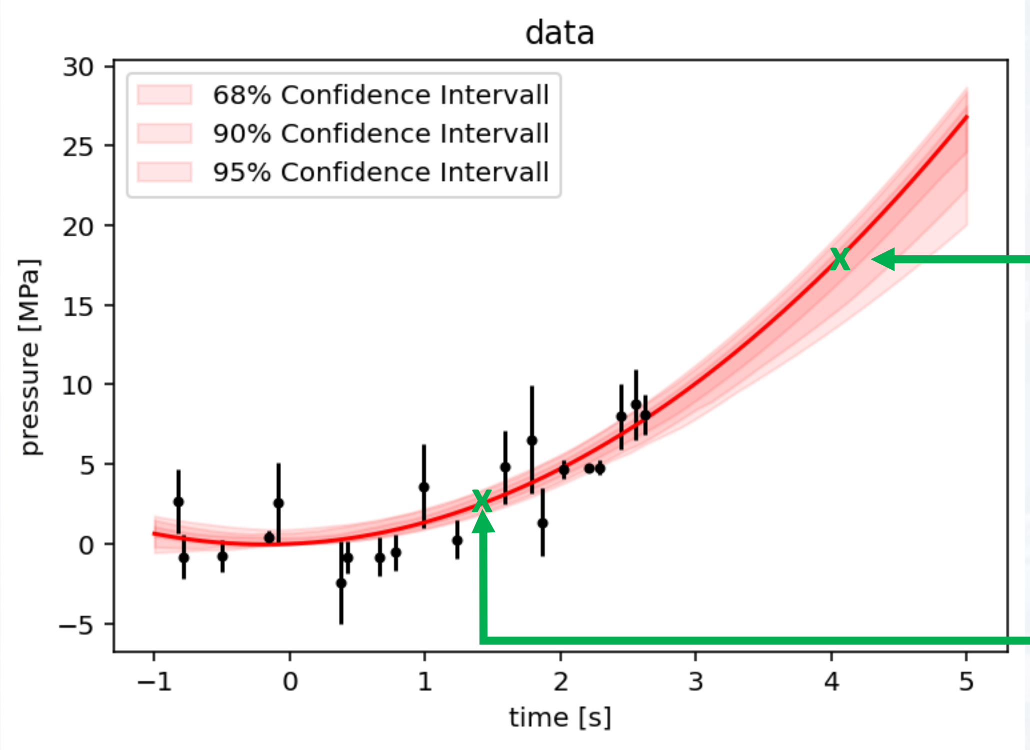
model: $f(x_i) = m\,x_i + n$
parameter: m, n



model: $f(x_i) = a\,x_i^2 + b\,x_i + c$
parameter: a, b, c

problem: set of data points $y_i = f(x_i)$, ideally each data point y_i has an error σ_i
we know $f(x_i)$

goal: find parameters of $f(x_i)$
once we know the parameters → **prediction**



y_i for data point ***outside*** known interval

y_i for data point ***within*** known interval

Data Fitting and Regression:

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we know $f(x_i)$

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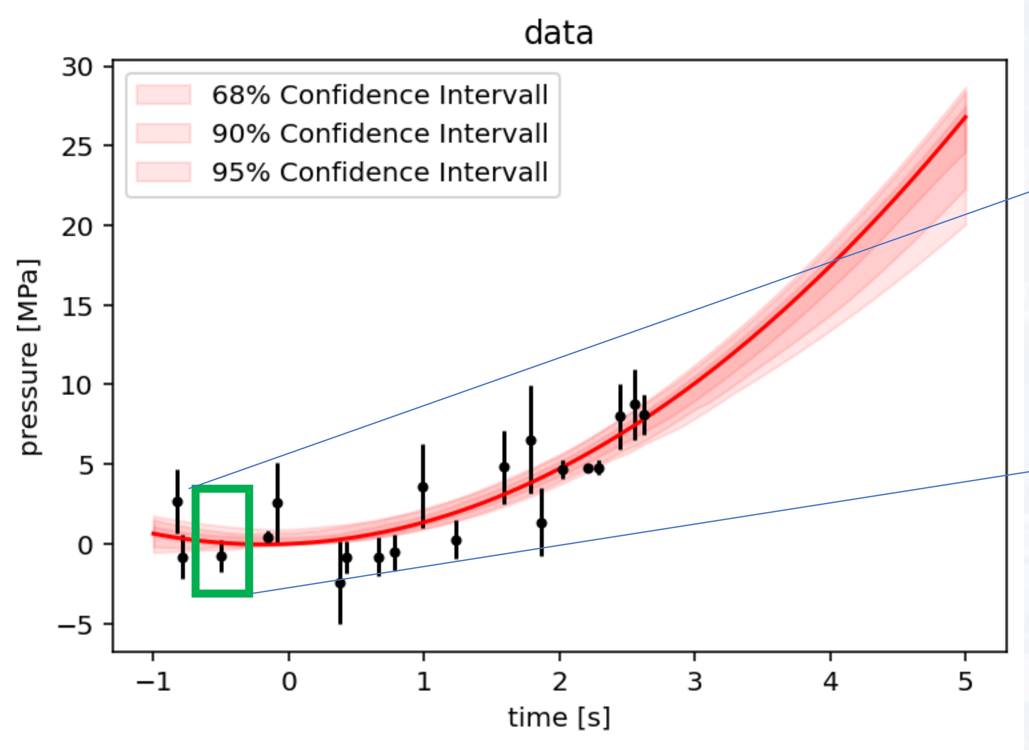
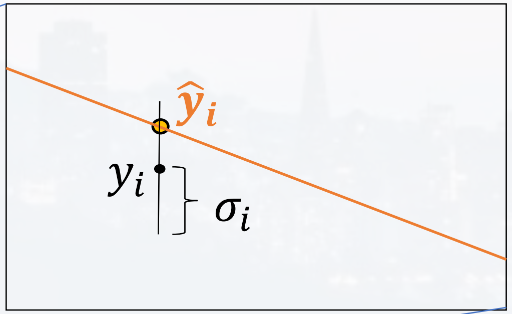
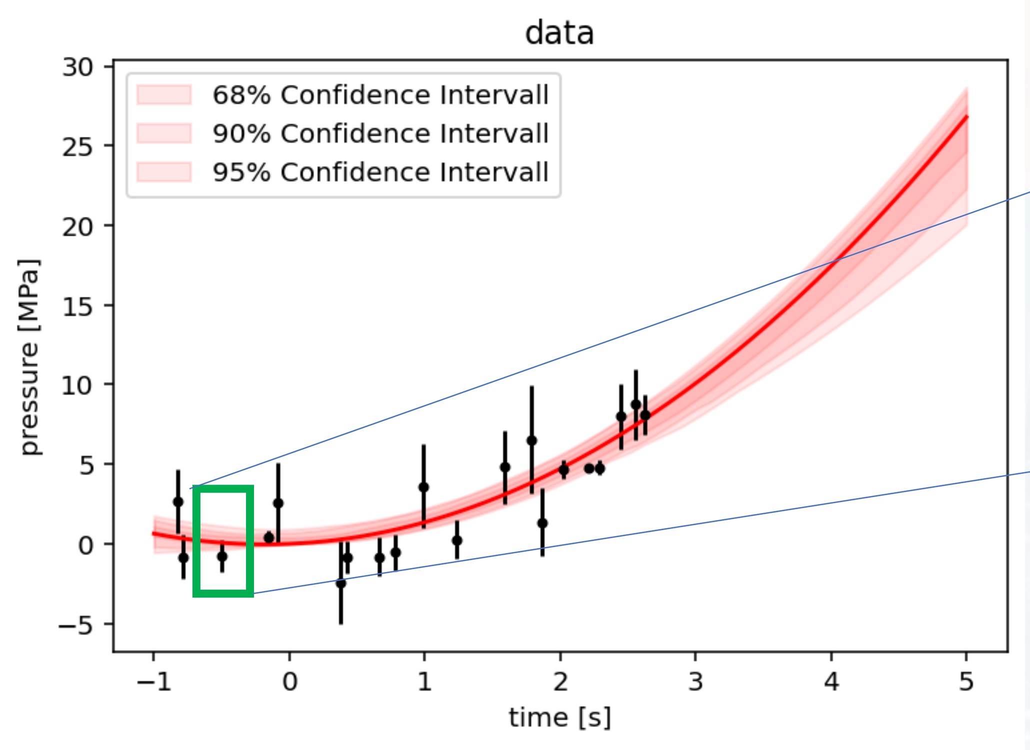
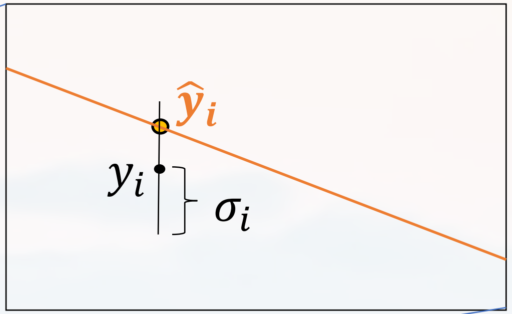




y_i: measured value of data point
σ_i: statistical error of y_i (often aka ey_i)
$\hat{y}_i$: prediction by the model *after the fit*
N : number of data points
p: number of fit parameter

Data Fitting and Regression:





finding best parameters by minimizing

$$\chi^2_{red} = \frac{1}{df}\sum_{i=1}^{N}\left(\frac{y_i - \hat{y}_i}{\sigma_i}\right)^2$$

$df = N - p - 1$

see module 8

or

$$MSE = \frac{1}{df}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

if no errors given

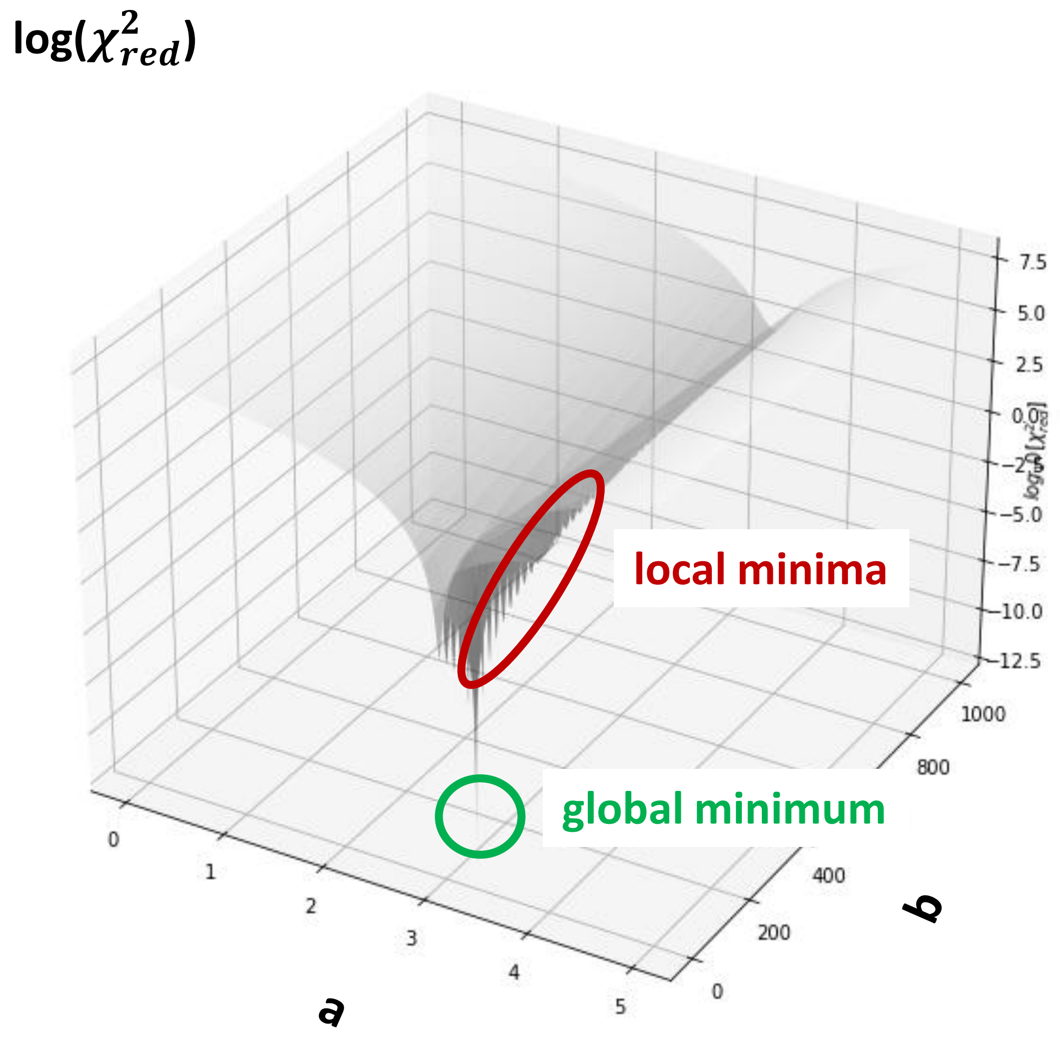
y_i: measured value of data point
σ_i: statistical error of y_i (often aka ey_i)
$\hat{y}_i$: prediction by the model *after the fit*
N : number of data points
p: number of fit parameter

finding best parameters by minimizing

$$\chi^2_{red} = \frac{1}{df}\sum_{i=1}^{N}\left(\frac{y_i - \hat{y}_i}{\sigma_i}\right)^2 \qquad df = N - p - 1$$

see module 8

or $$MSE = \frac{1}{df}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$ if no errors given

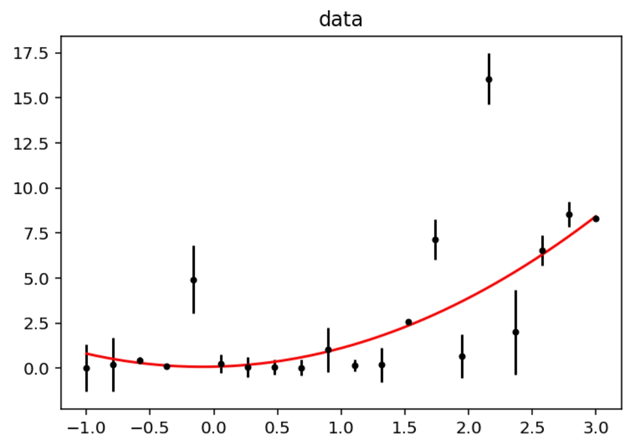


finding the global minimum of a higher dimensional non-analytical function

→ see also module 3

Data Fitting and Regression:

from module 8



$$\chi^2_{red} = \frac{1}{df}\sum_{i=1}^{N}\left(\frac{y_i - \hat{y}_i}{\sigma_i}\right)^2$$

$$df = N - p - 1$$

N : number of data points
p: number of fit parameter (model)

given a fitted model:
χ^2_{red} is a measure of the fit quality

for large **(> 50…100)** N **(number of data points)**:

≈ **2/3** of the data points should be consistent with the model within their **1σ** error bars

≈ **95%** of the data points should be consistent with the model within their **2σ** error bars

≈ **99.7%** of the data points should be consistent with the model within their **3σ** error bars

$\chi^2_{red} \approx$

1.0 excellent fit
1.0…1.5 acceptable fit
1.5…1.7 bad fit
>2.0 not acceptable

<<1.0 suspicious, errors are overestimated!

```
from scipy.optimize import curve_fit

def fun_to_fit(x, a, b, c):
        return a*x**2 + b*x + c

ValsBest, Cov = curve_fit(fun_to_fit, x, y)
```

```
ValsBest
array([ 1.37347967, -0.31706189, -0.20746552])
```

a b c

```
In [16]: Cov
Out[16]:
array([[ 0.15053344, -0.28309719, -0.05890134],
       [-0.28309719,  0.6630454 , -0.02965514],
       [-0.05890134, -0.02965514,  0.33825647]])
```

$$\begin{pmatrix} \sigma_a^2 & cov(a,b) & cov(a,c) \\ cov(b,a) & \sigma_b^2 & cov(b,c) \\ cov(c,a) & cov(c,b) & \sigma_c^2 \end{pmatrix}$$

Data Fitting and Regression:


```
from scipy.optimize import curve_fit
```

scipy has the optimization tool cuve_fit

```
def fun_to_fit(x, a, b, c):
        return a*x**2 + b*x + c
```

defining the model as a python function

```
ValsBest, Cov = curve_fit(fun_to_fit, x, y)
```

run the curve fit

```
ValsBest
array([ 1.37347967, -0.31706189, -0.20746552])
         a             b              c
```

best values by minimizing MSE or related

```
In [16]: Cov
Out[16]:
array([[ 0.15053344, -0.28309719, -0.05890134],
       [-0.28309719,  0.6630454 , -0.02965514],
       [-0.05890134, -0.02965514,  0.33825647]])
```

$$\begin{pmatrix} \sigma_a^2 & cov(a,b) & cov(a,c) \\ cov(b,a) & \sigma_b^2 & cov(b,c) \\ cov(c,a) & cov(c,b) & \sigma_c^2 \end{pmatrix}$$

covariance matrix for the model parameters

- **non diagonal values should be ≈ 0** (mutually independent)
- **diagonal: squared errors** (see module 8)

Data Fitting and Regression:


```
from scipy.optimize import curve_fit

def fun_to_fit(x, a, b, c):
        return a*x**2 + b*x + c

ValsBest, Cov = curve_fit(fun_to_fit, x, y)
```

$$a = 1.37 \pm 0.39$$
$$b = -0.32 \pm 0.81$$
$$c = -0.21 \pm 0.58$$

```
ValsBest
array([ 1.37347967, -0.31706189, -0.20746552])
```

a b c

```
In [16]: Cov
Out[16]:
array([[ 0.15053344, -0.28309719, -0.05890134],
       [-0.28309719,  0.6630454 , -0.02965514],
       [-0.05890134, -0.02965514,  0.33825647]])
```

$$\begin{pmatrix} \sigma_a^2 & cov(a,b) & cov(a,c) \\ cov(b,a) & \sigma_b^2 & cov(b,c) \\ cov(c,a) & cov(c,b) & \sigma_c^2 \end{pmatrix}$$

```
error_1sigma = np.sqrt(np.diagonal(Cov))
```

Data Fitting and Regression:

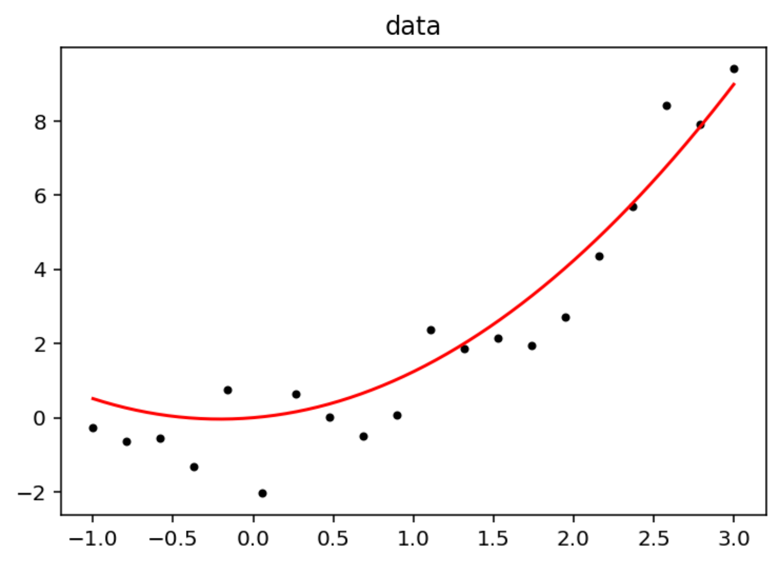

```python
from scipy.optimize import curve_fit

def fun_to_fit(x, a, b, c):
        return a*x**2 + b*x + c

ValsBest, Cov = curve_fit(fun_to_fit, x, y)
```

$$a = 1.37 \pm 0.39$$
$$b = -0.32 \pm 0.81$$
$$c = -0.21 \pm 0.58$$

$$y = 1.37\,x^2 - 0.32\,x - 0.21$$



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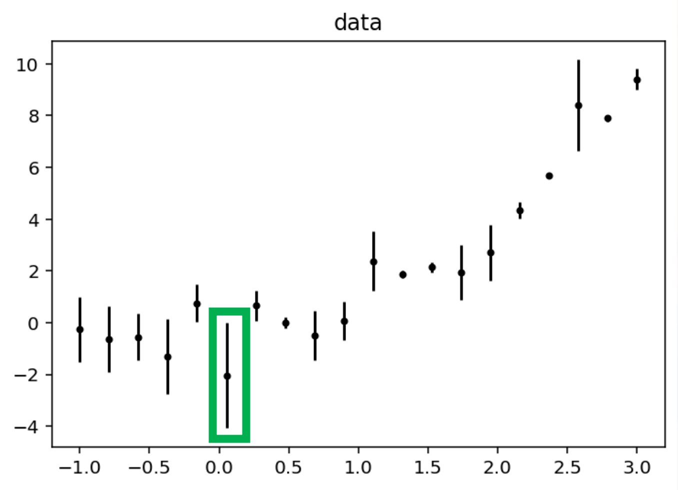
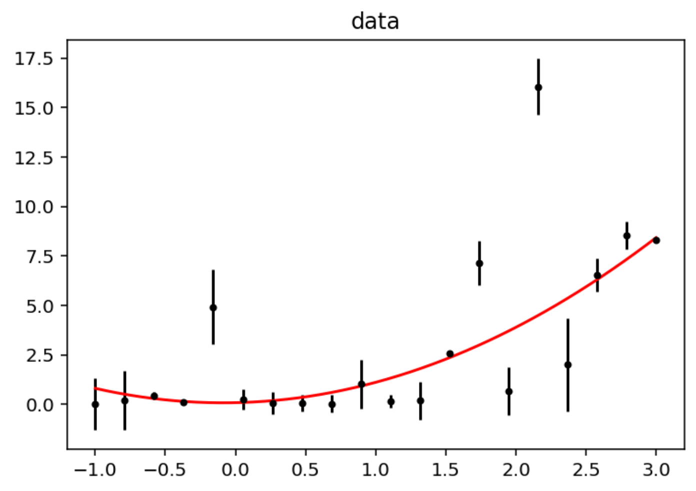
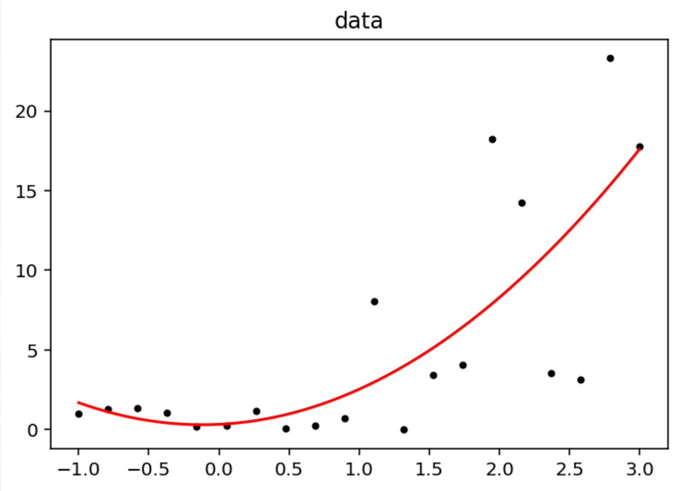


if error bars known → **error weighted fit!**

```
ValsBest, Cov = curve_fit(fun_to_fit, x, y, sigma = err, absolute_sigma = True)
```



vs

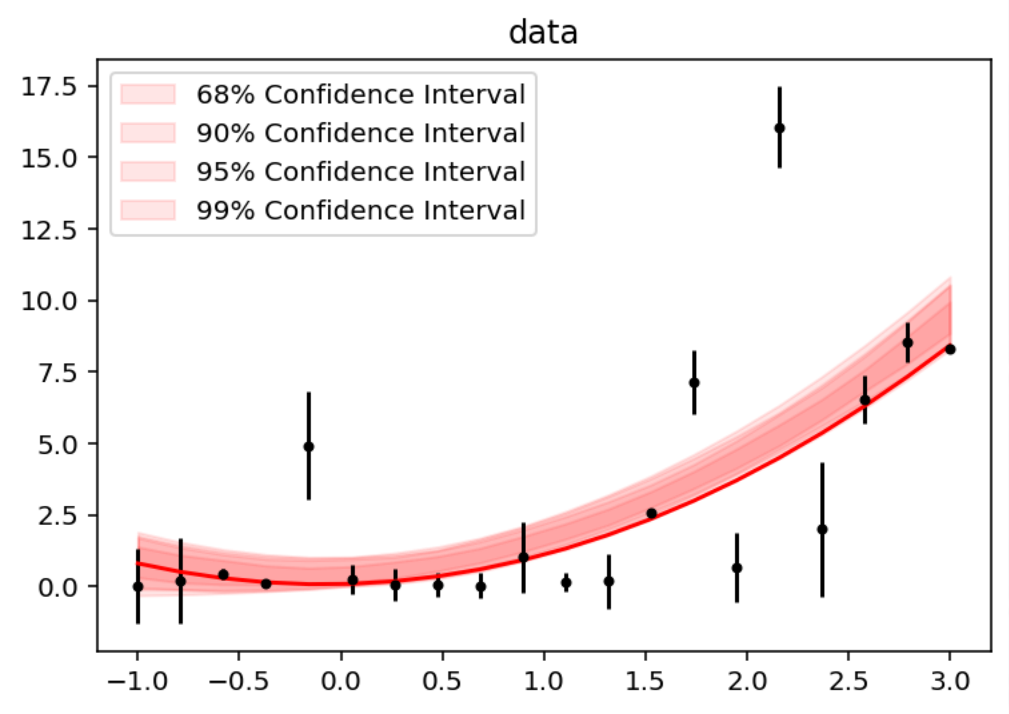


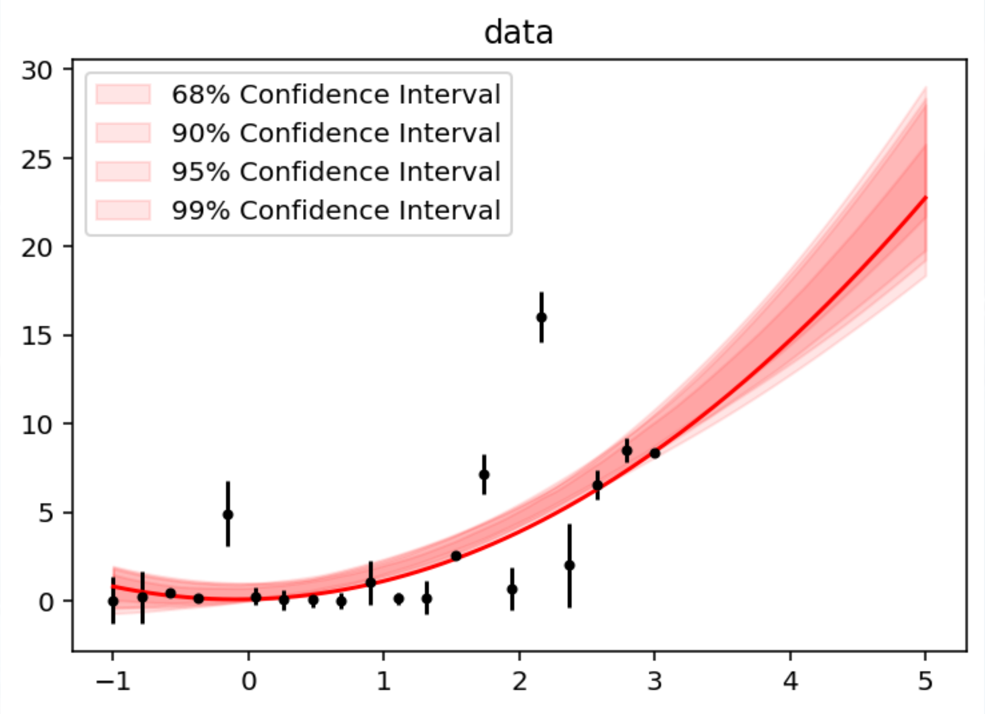
Data Fitting and Regression:

data points have errors and/or fit parameter have uncertainties too

$$\begin{aligned} a &= 1.37 \pm 0.39 \\ b &= -0.32 \pm 0.81 \\ c &= -0.21 \pm 0.58 \end{aligned}$$

→ implies uncertainties of the model
→ confidence band/interval





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→ implies uncertainties of the model
→ confidence band/interval
→ idea: **bootstrapping!**

$a = \;\; 1.37 \pm 0.39$
$b = -0.32 \pm 0.81$
$c = -0.21 \pm 0.58$

data points have errors and/or fit parameter have uncertainties too

→ implies uncertainties of the model
→ confidence band/interval
→ idea: **bootstrapping!**

if **no errors of y_i known**

→ assuming that fitted parameters follow a **normal distribution**, i.e. $a = 1.37 \pm 0.39$ where $\mu_a = 1.37$ and $\sigma_a = 0.39$ and so on…

→ **varying** the parameters **within their errors** using `np.random.normal(`μ_a`, `σ_a`, N)` **N** times

→ for each **N, generating a curve fit**

→ from set of N curve fits → **calculating percentiles** for confidence band/ interval

Data Fitting and Regression:

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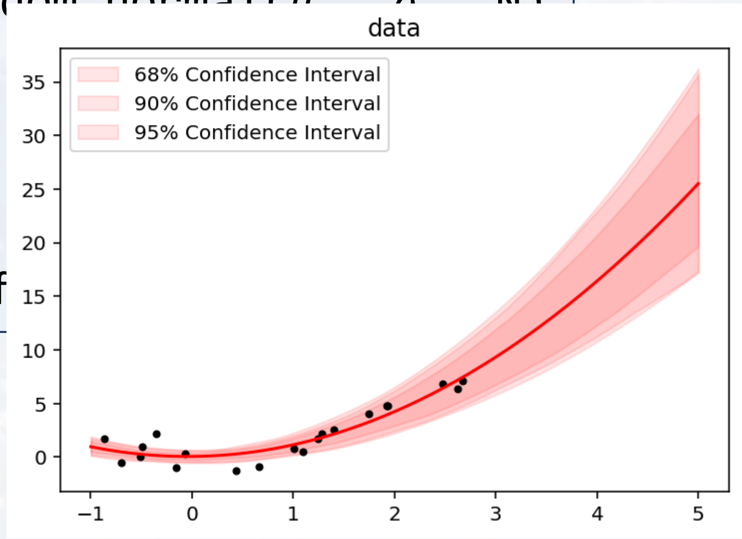
→ from set of N curve fits → **calculating percentiles** for conf



Data Fitting and Regression:

data points have errors and/or fit parameter have uncertainties too

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→ implies uncertainties of the model
→ confidence band/interval
→ idea: **bootstrapping!**

if **errors of y_i known**

→ assuming that errors of y_i follow a **normal distribution**, i.e. $\boldsymbol{y_i(boot) = y_i \pm \sigma_i}$

→ **varying** all y_i **within their errors** using `np.random.normal(`y_i`,` σ_i`, N)` **N** times

→ for each **N, generating a curve fit**

→ from set of N curve fits → **calculating percentiles** for confidence band/ interval data points have errors and/or fit parameter have uncertainties too

$$a = 1.37 \pm 0.39$$
$$b = -0.32 \pm 0.81$$
$$c = -0.21 \pm 0.58$$

→ implies uncertainties of the model
→ confidence band/interval
→ idea: **bootstrapping!**

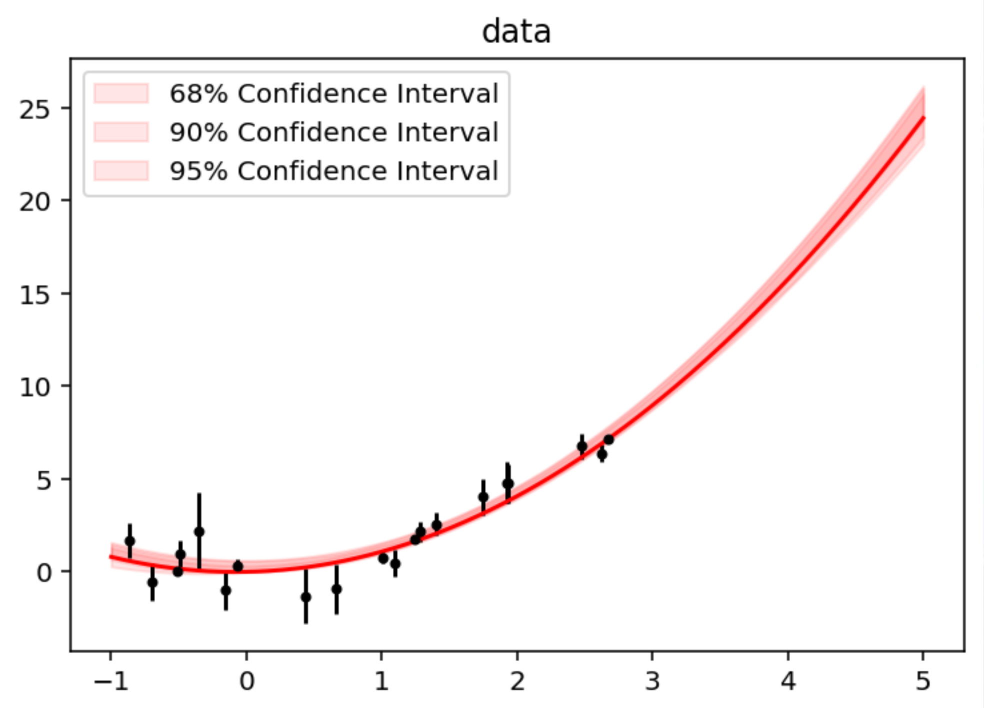
if **errors of y_i known**

→ assuming that errors of y_i follow a **normal distribution**, i.e. $y_i(boot) = y_i + \sigma_i$

→ **varying** all y_i **within their errors** using `np.ra`
N times

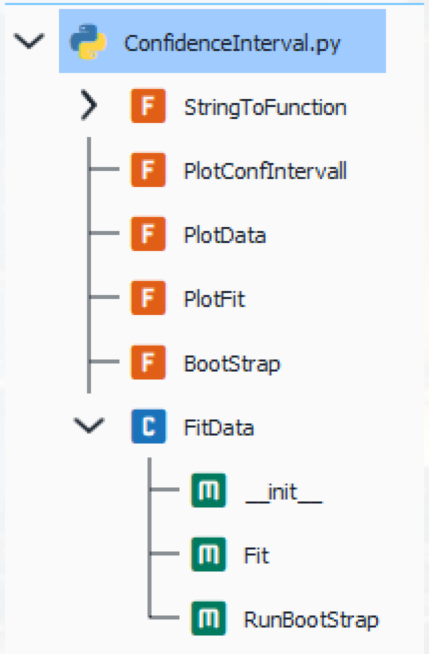
→ for each **N, generating a curve fit**

→ from set of N curve fits → **calculating percen**



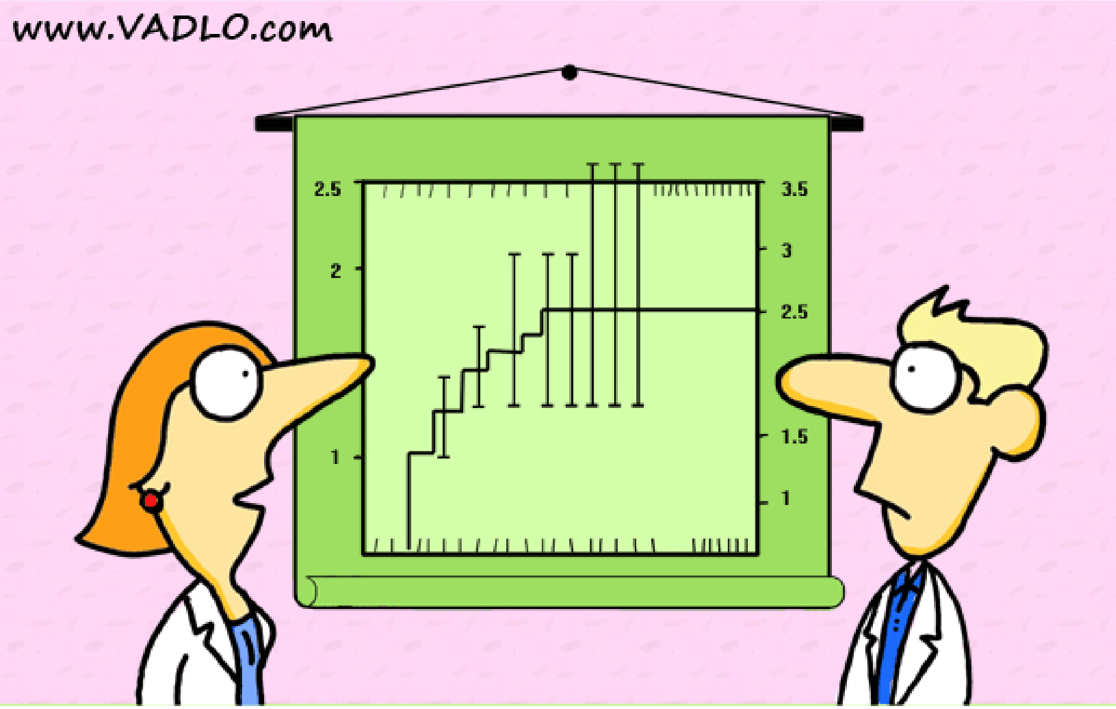
bootstrapping

explore the `.py` file
`ConfidenceInterval.py`




```
USAGE:
    generating a test sample:
    
    x         = np.linspace(-1,3,20)
    err       = np.random.normal(0, 1, (len(x),))#1sigma errorbars
    y         = x**2 + err
    errorbars = abs(err)
    
    
    1) plotting data
        
    F1 = FitData(x, y)
    F2 = FitData(x, y, errorbars)
    F3 = FitData(x, y, errorbars, time = '[s]', pressure = '[MPa]')
    
    
    2) fitting data (returns best values of fitted params, 1sigma confidence and
                     reduced chi2 if errorbars given, MSE else)
                     
    res1  = F1.Fit()
    res2  = F2.Fit()
    res2  = F3.Fit()
    
    res12 = F1.Fit("a*x**2", [1], (-0.5, 10))
    
    
    3) Bootstrapping (either varying within errorbars or within conf of fitted
                      params)
                      
    F1.RunBootStrap()
    F2.RunBootStrap()
    F3.RunBootStrap()
    
    F1.RunBootStrap(100, [90, 95], np.linspace(-1,5,200))
    F3.RunBootStrap(100, [90, 95], np.linspace(-1,5,200))
"""
```

Thank you very much for your attention!



“Did you really have to show the error bars?”